



TECHNICAL PROPOSAL

Version 1.0

Ref No. -MDS/2026/HB/HCT/07092026

SUBMITTED TO



IN RESPONSE TO RFP TITLED

“Secure Access Service Edge (SASE) – 3 years”

RFP# NE-26-0129

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1. ORGANIZATION OF THIS DOCUMENT

This section provides an overview of the sections of this document and how they combine to form a complete proposal.

This document contains the following sections:

- **Executive Summary**
Provides a summary of the proposal for the executive management.
- **Technical Proposal**
This section contains the details of MDS's deliverables. Its sub-sections may include the following:
 - The Proposed Design section provides the details of the proposed design (including sizing details, if available). This section may not be present for RFQ or tenders with pre-defined BOQs
 - The Bill of Materials section lists the proposed products and/or services. Moreover, this section will list the optional products/services if proposed.
 - The Professional Services section (if proposed) lists the scope and exclusions thereof.
 - The Post Implementation Support (SLA) section lists the included services and the terms thereof.
 - Assumptions and Important notes about the proposed solution
 - Client's responsibilities & Pre-requisites section listing tasks/activities which should be undertaken by the customer in addition to any items which should be arranged by the customer.
- **Proposed Value Add**
This section lists any value-added services/offers which are proposed to the customer as part of our offer.
- **Project Management & Governance**
This section describes our Project Management & Governance Methodology and may also include High Level Project Plan and Risks
- **MDS Profile**
This section provides a brief overview of our company
- **Our Partnerships**
List of our partnerships with principal vendors and OEMs
- **Appendices**
Contain various supplementary information e.g. Technical Specifications of the proposed products, Partnership Certificates, Profiles of Engineers, Compliance tables etc. as required or requested by the customer.

2. EXECUTIVE SUMMARY

MDS is pleased to present this proposal to Higher Colleges of Technology (hereafter referred to as HCT for brevity). This proposal constitutes our best offer to meet HCT's requirements. Products and/or services which comprise this solution, were chosen after careful analysis of your requirements and the shared information shared with us.

We are submitting this proposal in response to the Request for Proposal (RFP) NE-26-0129 for a comprehensive, cloud-delivered Secure Access Service Edge (SASE) solution. We have thoroughly reviewed your requirements and are confident that our industry-leading Prisma® SASE platform is uniquely positioned to meet and exceed HCT's strategic objectives for a unified, secure, and scalable network security architecture.

We understand HCT's vision to consolidate its fragmented security infrastructure, reduce operational complexity, and provide consistent, high-performance security for all users—staff, faculty, students, and contractors—across all 17 campuses and remote locations. Your goal to unify security for both managed (institution-owned) and BYOD (personally-owned) devices under a single, cloud-delivered service aligns perfectly with the core principles of our Prisma SASE solution.

Palo Alto Networks is the global cybersecurity leader, pioneering the SASE category and consistently recognized by industry analysts for our vision and execution. Our proposal details how Prisma SASE, a convergence of our best-in-class Prisma Access and Prisma SD-WAN solutions, delivers:

- **Unified, Cloud-Native Security:** A single platform that integrates Firewall as a Service (FWaaS), Secure Web Gateway (SWG), Zero Trust Network Access (ZTNA 2.0), and Cloud Access Security Broker (CASB) to eliminate security gaps and reduce complexity.
- **AI-Powered Threat Prevention:** Superior security powered by Precision AI™ to stop zero-day threats, advanced malware, and phishing attacks in real-time, ensuring HCT's data, users, and resources are protected against the most sophisticated cyberattacks.
- **Exceptional User Experience:** A massively-scaled, low-latency global network that ensures fast and secure access to all applications, whether in the cloud or the data center, from any location.
- **Simplified Operations:** A single, intuitive management console that provides centralized visibility, policy control, and AI-driven operational insights, significantly reducing the administrative burden on HCT's IT team.

This proposal provides a detailed response to all sections of your RFP, outlining our technical capabilities, project methodology, support services, and commercial structure. We are also pleased to offer a comprehensive Proof of Concept (PoC) to demonstrate the superior performance and security of our solution in HCT's live environment.

We are excited about the opportunity to partner with Higher Colleges of Technology and support your mission to be a leader in applied education. We look forward to discussing our proposal with you in further detail.

The MDS and Paloalto implementation teams is keen to deliver top class services in order to meet HCT's requirements.

We believe that a combination of expertise and professionalism of all project team members, from both sides, working together on the engagement will enable achieving the objectives stated above.

MDS is fully committed to be a strategic supplier of quality services to HCT and to bring the highest possible value to HCT by our:

- Commitment to quality and client satisfaction
- Knowledge of the existing core Infrastructure products
- Understanding of Client's needs and requirements

We strongly believe that HCT will benefit from the project outputs in the long-term perspective, and that these outputs will staunchly support its strategic business needs and objectives.

Sincerely,

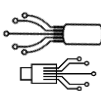
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3. TECHNICAL PROPOSAL

3.1 MDS'S RESPONSIBILITIES

MDS's technical team has put its best efforts towards understanding your requirements and formulating this proposal. Based on this understanding our offer includes the following:

	Supplying the Product	✓
	MDS Professional Services:	
	<i>MDS Plan, Design, Implement Services</i>	✓
	<i>Vendor Professional Services</i>	✓
	<i>Cabling & DC Room related Services</i>	✓
	<i>DC Room, Racks, Power & UPS Services</i>	⊘
	<i>Lift & Shift Services</i>	✓
	<i>Knowledge Transfer Services</i>	✓
	Post Implementation Annual Support Services	✓
Project Management & Governance		

3.2 PROPOSED DESIGN

Higher Colleges of Technology (HCT) is at a pivotal point in its digital transformation, seeking to modernize its network security infrastructure to support a dynamic and distributed academic environment. The current fragmented approach, with separate on-premises and cloud solutions for different user populations, creates operational complexity, inconsistent policy enforcement, and visibility gaps. HCT requires a unified, fully cloud-based Secure Access Service Edge (SASE) platform to consolidate internet traffic security for all users (staff, faculty, students, contractors) and devices (managed and BYOD) across its 17 campuses and data centers.

Palo Alto Networks proposes **Prisma® SASE**, the industry's most complete SASE solution, to address HCT's challenges and achieve its strategic objectives. Our proposal is built on the understanding that HCT needs a partner who can deliver not just a product, but a comprehensive security transformation that is scalable, agile, and future-proof. Our proven expertise in this area is highlighted by successful customer transformations for organizations with similar complex requirements, such as the Ministry of Justice, whose network and security transformation with Palo Alto Networks is detailed in their case study: <https://www.paloaltonetworks.co.uk/customers/mojs-network-and-security-transformation-with-palo-alto-networks>.

Figure 1: Prisma SASE's unified, cloud-delivered architecture secures diverse users and locations accessing cloud and data center resources, leveraging Zero Trust principles, AI Access Security, and Enterprise DLP for comprehensive network and security transformation.

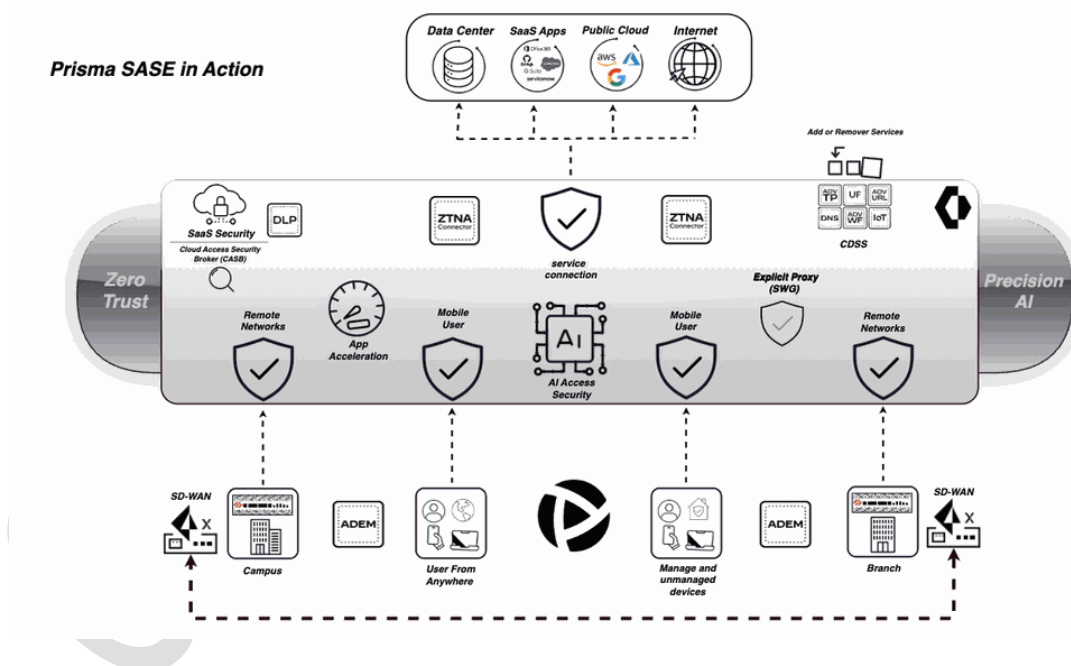


Figure 1: Prisma SASE's unified, cloud-delivered architecture secures diverse users and locations accessing cloud and data center resources, leveraging Zero Trust principles, AI Access Security, and Enterprise DLP for comprehensive network and security transformation.

Our Understanding of HCT's Key Objectives:

- **Unified Internet Security:** Consolidate disparate security tools into a single, cloud-delivered platform.
- **Enhanced Security:** Leverage AI-powered threat prevention and real-time intelligence to protect against modern cyberthreats.
- **User and Device Parity:** Enforce consistent security policies for all users and devices, regardless of location.
- **Cloud Optimization:** Enable secure, high-performance access to cloud applications and services.
- **Zero Trust Integration:** Build a comprehensive Zero Trust security model that integrates with HCT's existing ecosystem.
- **Simplified Management:** Reduce operational overhead with a centralized, intuitive management console.
- **Scalability:** Implement a solution that can grow with HCT's needs, including new campuses and increased cloud adoption.

The Palo Alto Networks Solution: Prisma SASE

Prisma SASE is the convergence of networking and security into a single, cloud-native platform that secures HCT's hybrid workforce and prepares the institution for its cloud-first future. Our solution directly addresses HCT's requirements by delivering:

- **A Unified, Cloud-Native Platform:** Prisma SASE integrates best-in-class security capabilities, including **Prisma Access** (our Security Service Edge - SSE), which unifies Firewall as a Service (FWaaS), Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), and the industry's only ZTNA 2.0. This consolidation eliminates the need for multiple point products, reduces complexity, and closes security gaps.

Figure 2: This diagram illustrates how Prisma Access unifies core security services like FWaaS, SWG, CASB, DLP, and ZTNA 2.0 into a single, AI-powered cloud platform, eliminating point products and closing security gaps.

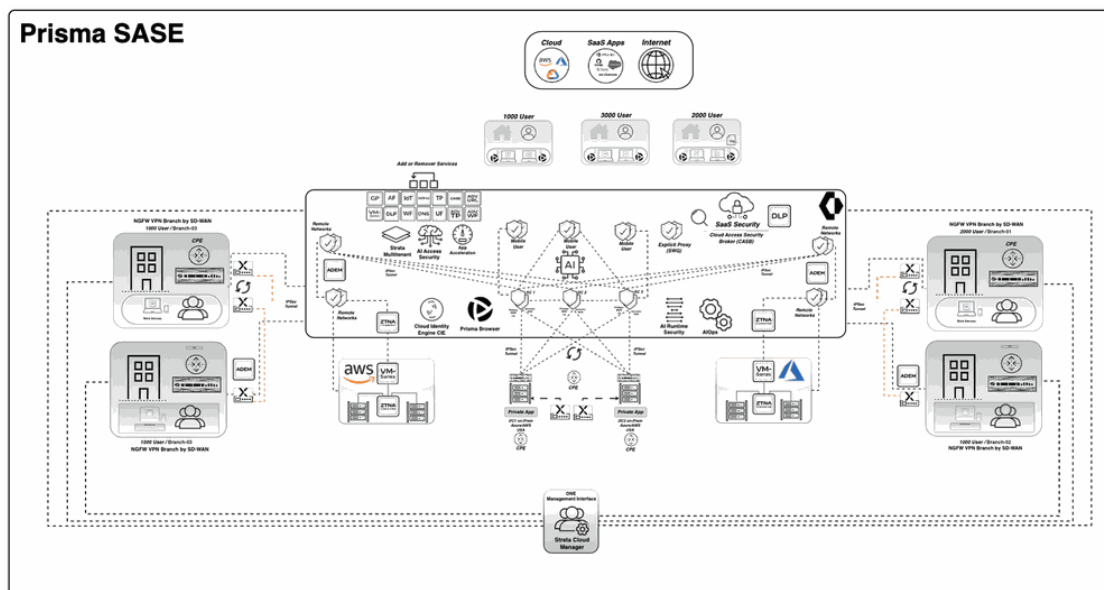


Figure 2: This diagram illustrates how Prisma Access unifies core security services like FWaaS, SWG, CASB, DLP, and ZTNA 2.0 into a single, AI-powered cloud platform, eliminating point products and closing security gaps.

- **Superior AI-Powered Security:** Our solution is powered by **Precision AI™**, leveraging deep learning and machine learning models trained on the industry's largest dataset to stop zero-day threats in real-time. This includes advanced phishing detection, zero-day malware prevention, and granular controls for Generative AI applications, ensuring HCT is protected from the most sophisticated cyberattacks.
- **Consistent Policy for All Users and Devices:** We provide a unified policy framework for both managed devices (via the lightweight **Prisma Access Agent**) and BYOD devices (via the agentless **Prisma Browser**). This ensures every user, from faculty to students and contractors, receives the same high level of security, regardless of their device or location.

Figure 3: Prisma Access Agent on managed devices and agentless Prisma Browser for BYOD provide consistent, secure access to all applications, including AI Apps, through Prisma SASE. This architecture ensures unified policy enforcement and advanced security, like Enterprise DLP and threat prevention, regardless of user device or location.

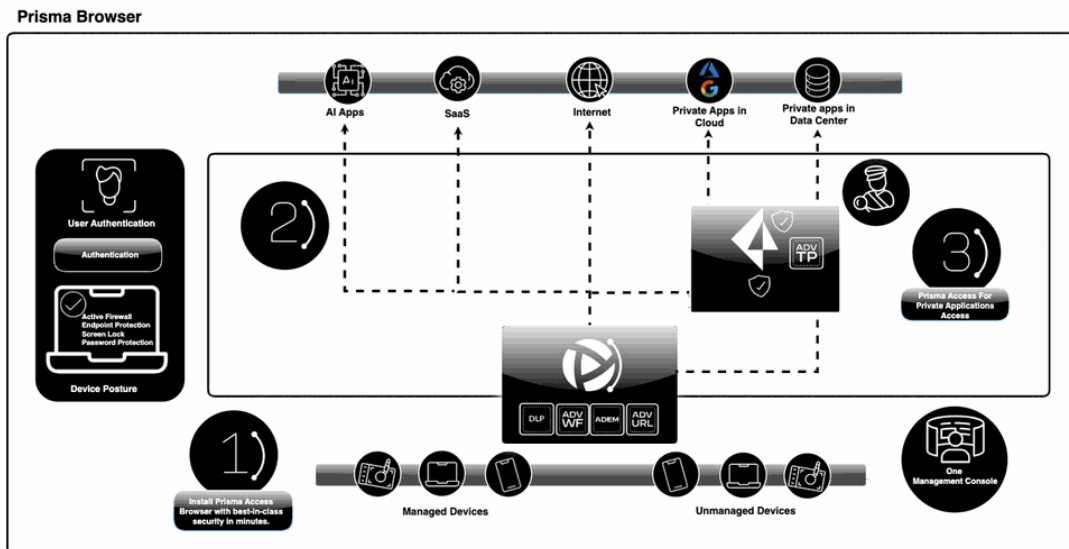


Figure 3: Prisma Access Agent on managed devices and agentless Prisma Browser for BYOD provide consistent, secure access to all applications, including AI Apps, through Prisma SASE. This architecture ensures unified policy enforcement and advanced security, like Enterprise DLP and threat prevention, regardless of user device or location.

- Simplified Operations and Management: Strata Cloud Manager** provides a single pane of glass for managing the entire SASE fabric. Its AI-powered operations (AIOps) and natural-language Copilot simplify troubleshooting, automate policy recommendations, and provide deep analytics, empowering HCT's IT team to manage security more effectively and efficiently.

Figure 4: Strata Cloud Manager provides a unified, single pane of glass for centralized Prisma SASE management, offering deep analytics and comprehensive control over security policies, including Enterprise DLP, to empower IT teams.

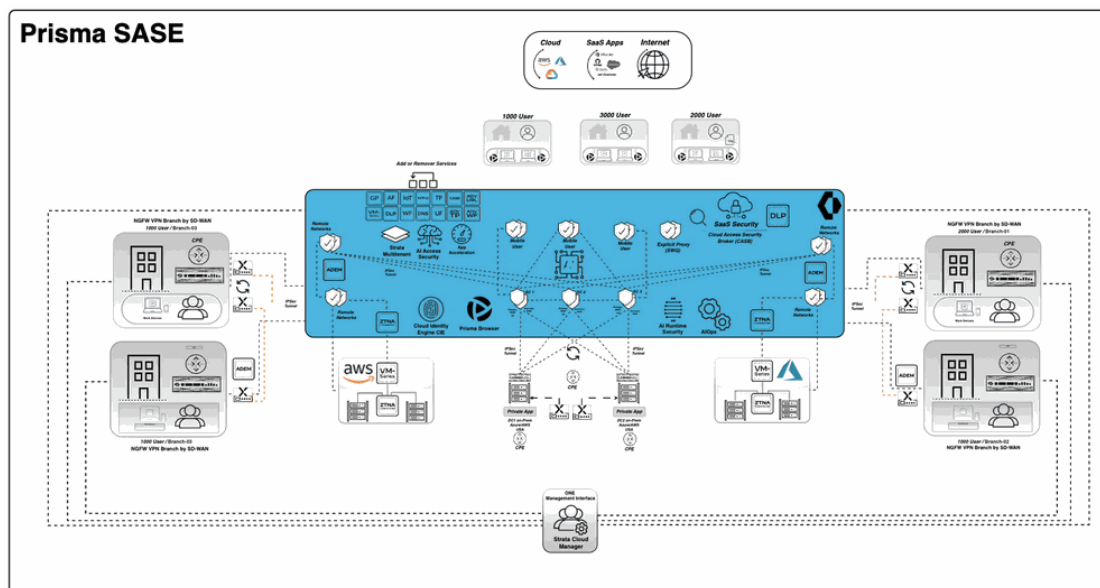


Figure 4: Strata Cloud Manager provides a unified, single pane of glass for centralized Prisma SASE management, offering deep analytics and comprehensive control over security policies, including Enterprise DLP, to empower IT teams.

- **Exceptional User Experience:** With over 150 global Points of Presence (PoPs), our massively scaled network ensures low-latency, high-performance access to applications. Our integrated **Autonomous Digital Experience Management (ADEM)** provides end-to-end visibility, proactively identifying and resolving issues before they impact users.
- **Optional ZTNA for Private Applications:** Our proposal includes a fully-costed optional scope for ZTNA 2.0, providing secure, identity-aware, application-level access to on-premises and data center resources for 200 designated users, completely replacing legacy VPNs with a superior, more secure architecture.

Palo Alto Networks is not just a vendor; we are a strategic partner committed to HCT's success. Our proposal includes comprehensive professional services for a seamless migration, dedicated support, and in-depth training to ensure HCT's team is fully equipped to manage the new platform. By choosing Palo Alto Networks, HCT is investing in a future-proof security architecture that will scale with its growth, protect its community from evolving threats, and empower its mission of educational excellence.

3.3 BILL OF MATERIALS

The section lists the Bill of Materials for the proposed products, including any preferred and/or optional products as mentioned in the executive summary.

3.3.1 MAIN OFFER OPTION (1): SASE - PRISMA BROWSER PRO (3 YEAR PREM SUPPORT).

#	Part Number	Description	Qty
Option (1): SASE - Prisma Browser Pro (3 Year PREM Support).			
1	PAN-PRISMA-BROWSER-PRO	Prisma Browser Pro includes Enterprise Browser for Internet/SaaS/Private app access, SWG capabilities, Enterprise DLP, AI Access, Management Console, Adv. Web, Data, and Identity Protection, ADEM RUM, and Standard Support per user/year	29022
2	PAN-PRISMA-ACCESS-STDALNSEB-PREM	Premium Support for Prisma Access Secure Enterprise Browser Standalone	1
3	PS Outcome-Based	PAN-CONSULT-CUSTOM-PATHis is purely prepaid service and billing will be done upon receipt of PO without any POD	1
4	PAN-PA-5410-AC	Palo Alto Networks PA-5410 with redundant AC power supplies	3
5	PAN-PA-5410-BND-PRECISIONAI-3YR	PA-5410, Precision AI Network Security Subscription Bundle (Advanced Threat Prevention, Advanced URL Filtering, Advanced Wildfire, Advanced DNS Security, Advanced SD-WAN and Device Security), 3 years (36 months) term	3
6	PAN-PA-5410-PAA-3YR	PA-5410, Prisma Access Agent subscription, 3 years (36 months) term.	3
7	PAN-SVC-PREM-5410-3YR	Premium support 3-year term, PA-5410	3
8	PAN-PWR-CORD-UK	Power cord for United Kingdom with IEC-60320 C13 and BS 1363 UK13 cord ends, 10A, 250V max, 6ft	6
9	PAN-QSFP-AOC-10M	QSFP+ form factor, 40Gb active optical cable with 2 transceivers and 10m of cable permanently bonded as an assembly	2
10	PAN-SFP-PLUS-SR	SFP+ form factor, SR 10Gb optical transceiver, short reach 300m, OM3 MMF, duplex LC, IEEE 802.3ae 10GBASE-SR compliant	9
11	PAN-SFP-PLUS-SR	SFP+ form factor, SR 10Gb optical transceiver, short reach 300m, OM3 MMF, duplex LC, IEEE 802.3ae 10GBASE-SR compliant	9
12	PAN-SFP28-25GBASE-SR	SFP28 form factor, SR 25Gb optical transceiver, short reach 100m, MMF, duplex LC, IEEE 802.3ae 25GBASE-SR compliant	9
13	PAN-QSFP-40GBASE-BIDI	QSFP+ form factor, 40Gb Bidirectional optical transceiver, 100m reach over OM3 MMF, 150m over OM4 MMF, duplex LC	9
14	PAN-SFP28-25GBASE-LR	SFP28 form factor, LR 25Gb optical transceiver, long reach 10Km, SMF, duplex LC, IEEE 802.3ca 25GBASE-LR compliant	9
15	PAN-SFP-PLUS-LR	SFP+ form factor, LR 10Gb optical transceiver, long reach 10Km, SMF, duplex LC, IEEE 802.3ae 10GBASE-LR compliant	9
16	PAN-QSFP-40GBASE-LR4	QSFP+ form factor, 40Gb LR4 optical transceiver, long reach 10Km, SMF, duplex LC, IEEE 802.3ba 40GBASE-LR4 compliant	9

Services			
1	Annual Support Contract	3 Year Partner Support Services as per the below: - 4 On-Site Visits per year - Unlimited Phone/Email Support - Health Checks = 4 per year - Hardware Replacement = Back-to-Back with the Vendor (provided affected HW is under support) - SLA Response time as per the details of the proposal	1

3.3.2 OPTION (2): SASE - PRISMA ACCESS - CASB (3 YEAR PREM SUPPORT).

#	Part Number	Description	Qty
Option (2): SASE - Prisma Access - CASB (3 Year PREM Support).			
1	PAN-PRISMA-ACCESS-MU-ENTERPRISE	Prisma Access User Enterprise Edition includes private app access and outbound internet security for users, premium security subscription, worldwide service locations, 5 service connections, 10 ZTNA connectors ,and standard success per user per year	29022
2	PAN-PRISMA-ACCESS-PREM-SUCCESS	Prisma Access Premium Support	1
3	PAN-PRISMA-ACCESS-CASB	Prisma Access NG-CASB add-on, includes SaaS Security Inline, DLP In-line, SaaS Security API, DLP API, SSPM and AI Access Security, Standard Success plan per unit per year	29022
4	PAN-PRISMA-ACCESS-SEB-UPG	Prisma Browser Add-on to Prisma Access includes the Secure Enterprise Browser for Internet/SaaS and private app access, Enterprise DLP, AI Access security for PAB traffic only, Management Console, and Standard Success Plan per unit per year.	29022
5	PAN-PRISMA-ACCESS-MU-SLS	Strata Logging Service subscription for Prisma Access Mobile Users. Includes 1 year of retention. Price per user.	29022
6	PAN-PRISMA-ACCESS-SEB-UPG-PRO	Prisma Browser Pro Add-on to Prisma Access includes Enterprise Browser for Internet/SaaS/Private app access, Enterprise DLP, AI Access, Management Console, Advanced Web, Data, and Identity Protection, ADEM RUM per unit per year	29022
7	PAN-PRISMA-ACCESS-PRVLGED-RA	Prisma Access Secure Agentless Access (SAA) add-on includes access to non-web apps such as RDP/SSH/VNC and Standard Customer Success plan per user per year	2719
8	PS Outcome-Based	PAN-CONSULT-CUSTOM-PATHis is purely prepaid service and billing will be done upon receipt of PO without any POD	1

Services			
1	Annual Support Contract	3 Year Partner Support Services as per the below: - 4 On-Site Visits per year - Unlimited Phone/Email Support - Health Checks = 4 per year - Hardware Replacement = Back-to-Back with the Vendor (provided affected HW is under support) - SLA Response time as per the details of the proposal	1

3.3.3 EXCLUDED ITEMS

Any Items which cannot be quoted or offered should be mentioned here

3.3.4 ASSUMPTIONS/NOTES

This section lists all the assumptions and notices based on which this offer has been submitted.

- HCT should validate the proposed offer including the Bill of Quantities and the Scope of Work to ensure that it is aligned with their requirements before awarding the project.
- During initial requirement analysis thorough validation of HCT requirements has been done by us based on the information provided by HCT. This requirements has been translated into the Bill of Materials presented in the preceding section. Any changes identified after submission of this proposal will be considered a variation in the scope requiring a revised proposal or variation order depending upon the phase of the project.

3.4 PROPOSED ANNUAL SUPPORT SERVICES

This section lists the terms & conditions of the post implementation support services offered to HCT as part of this offer.

These terms & conditions will form the basis of the Support Services/SLA Contract at the time of award of the project.

3.4.1 TERMS OF SERVICES

1. Term of Support Contract = 3 Years
2. Coverage = 24x7x365
3. Access to Central Helpdesk for Problem Reporting and Ticket Logging during working hours
4. Dedicated Point of Contact for after working hours support
5. Health Checks per year = 2
6. OS Patching & Upgrades per year = 1
7. On Demand Reports of logged tickets

3.4.2 ITEMS COVERED

This SLA will cover the below mentioned items:

- Items which are proposed in this proposal are covered unless specified otherwise

3.4.3 LOCATIONS COVERED

Services will be delivered to HCT remotely and/or on-site at the following location(s):

- HCT's Location A

3.4.4 EXCLUDED SERVICES

This SLA doesn't cover below:

- Attending to, installing, configuring, or troubleshooting equipment not covered in this SLA.
- Provision of Spare equipment is not covered under this contract.
- Implementing new projects is not covered as part of this contract
- Installation and/or configuration of additional or new equipment (e.g. new switches, IP Phones, firewalls, etc.) which is not part of this project is not covered under this contract

- Being part of a change activity by any HCT department (e.g. Presence of networking resources during database or application migration/upgrade) is not covered under this contract.
- Major design changes or major configuration changes are not covered under this contract and should be initiated as “request fulfillment” with our Helpdesk and may be chargeable to HCT depending upon the nature of the request.

3.4.5 SERVICE LEVEL AGREEMENT

The following table describes the applicable Service Level Agreement for the Annual support contract.

Severity	Definition	Response Time ¹	Coverage
Critical	Situation affects entire enterprise and prevents further work	4 hours	24x7
High	Services severely degraded, but users can continue working with difficulties.	8 hours	24x7
Average	Working with the problem is inconvenient, but possible	1 Business Day (Sunday to Thursday)	8:00 am - 5:00 pm
Low	Situation is commonplace within the normal course of business.	5 Business days (Sunday to Thursday)	8:00 am - 5:00 pm

Important Notes:

- **Response time** is defined as the time required to respond to a ticket.
- The response time is measured from the time the support case is opened until the time the first contact is made with the HCT (via any communication medium such as email/phone/onsite visit/Fax etc.)
- Upon opening the incident ticket, HCT should provide full details of the issue supporting the severity claim to be assigned
- The response shall include troubleshooting or configurations on the client side as per the included services. Hardware replacements will be covered back-to-back by vendor through warranties. Any

¹ + Travel time in case the engineer has to travel to the site

corrective actions pertaining to non-covered equipment (e.g., servers, applications, 3rd party devices) should be performed by HCT.

- **Resolution of an incident** depends upon many factors some which may not be in our control, e.g., an unknown or zero-day bug without any bug fix from the vendor. Best efforts will be put to resolve an incident as per assigned priority however, no resolution time is being committed as part of this contract.

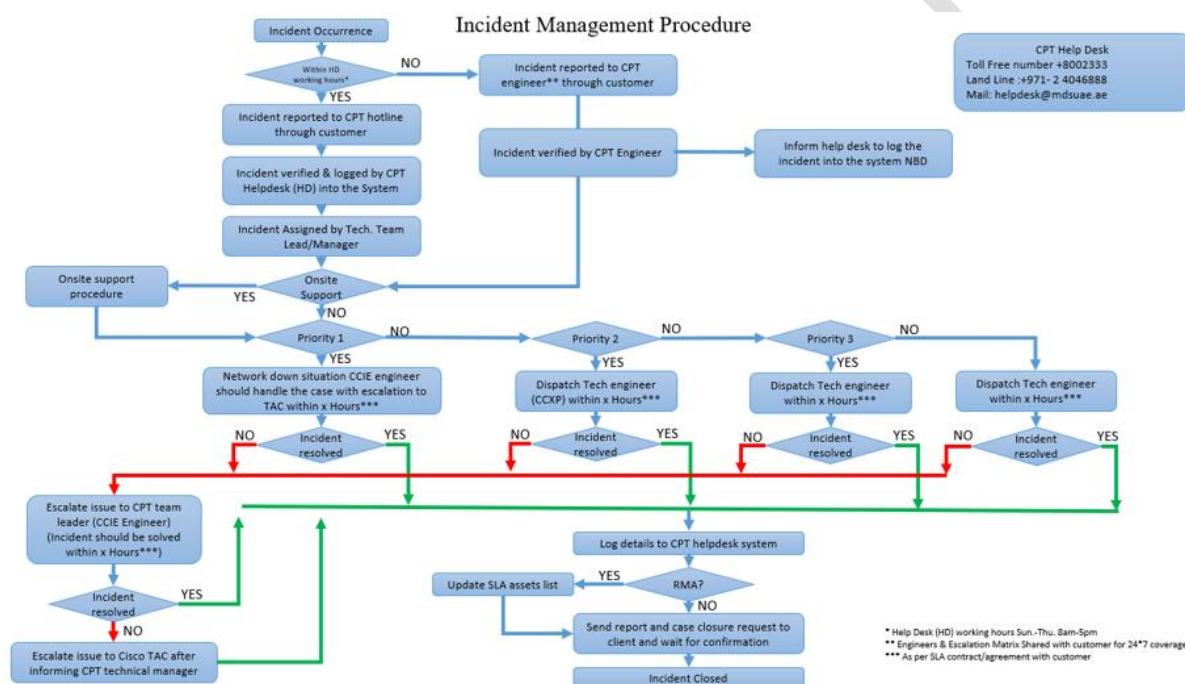
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3.4.6 FUNCTIONS OF THE HELPDESK

MDS has a central Helpdesk which handles all your support calls and keeps track of the SLA terms & conditions. This is the single point for requesting support. This Helpdesk is manned by professional Customer Service Representatives and provides the following functions:

3.4.6.1 Incident Management

The following process flow describes the lifecycle of an incident including severity levels and escalations:



3.4.6.2 Problem Management

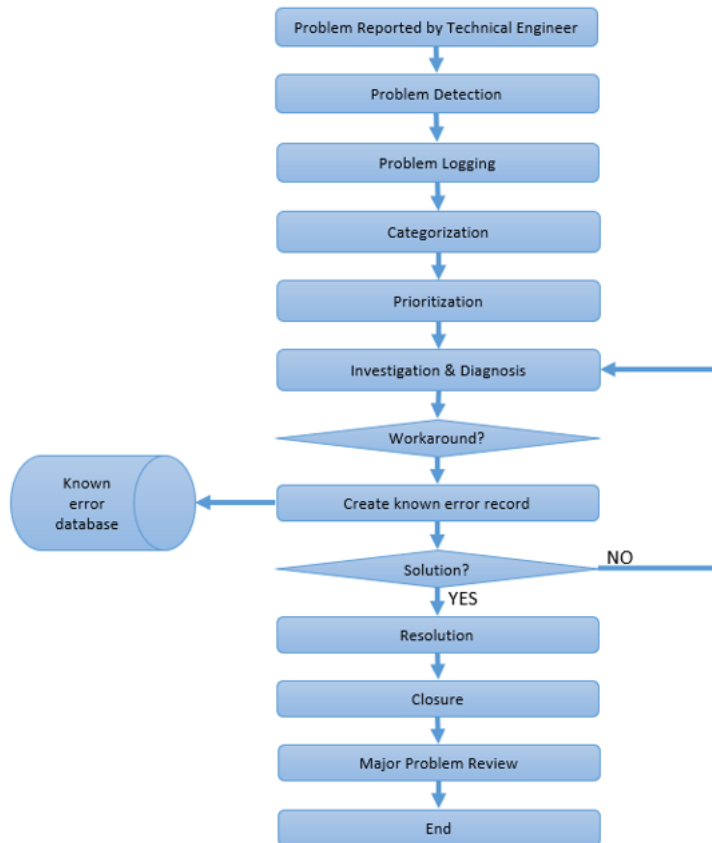
Problems are Interruptions in service identified by multiple incidents with the same symptoms or a single incident, and with unknown causes.

The primary objective of problem management is to prevent incidents from happening and to minimize the impact of incidents that cannot be prevented. Proactive problem management analyzes incident records, and uses data collected by other Service Manager processes to identify trends or significant problems.

Known errors: Problems for which the root causes have been identified and a workaround or a permanent fix has been identified, although the implementation of the permanent fix could be in the future.

The following diagram illustrates the lifecycle of a problem.

Problem Management Procedure



3.4.6.3 Automatic Escalation

When a ticket is logged, and a severity is assigned, the system automatically escalates the case if the ticket remains open and a threshold is reached as described in the escalation process through emails. Each opened ticket is monitored by the Helpdesk system as per the assigned classification and is escalated automatically as the SLA is lapsed based on the table below until it is closed.

Severity Definition	SEVERITY 1 Severity 1 means an existing Network or Environment is down or there is a critical impact to End User's business operation. End User and Cisco both will commit full-time resources to resolve the situation.	SEVERITY 2 Severity 2 means operation of an existing Network or Environment is severely degraded or significant aspects of End User's business operation are negatively impacted by unacceptable Network or Environment performance. End User and Cisco both will commit full-time resources during Standard Business Hours to resolve the situation.	SEVERITY 3 Severity 3 means operational performance of the Network or Environment is impaired, although most business operations remain functional. End User and Cisco both are willing to commit resources during Standard Business Hours to restore service to satisfactory levels.	SEVERITY 4 Severity 4 means information is required on Cisco product capabilities, installation, or configuration. There is little or no impact to End User's business operation. End User and Cisco both are willing to provide resources during Standard Business Hours to provide information or assistance as requested.
Priority Response Time	Highest	High	Normal	Low
	T0 25% 50% 75% 100%	T0 25% 50% 75% 100%	T0 25% 50% 75% 100%	T0 25% 50% 75% 100%
Alert Customer	✓	✓	✓	✓
Alert L1 Engineer	✓	✓	✓	✓
Alert Technical Manager	✓	✓	✓	✓
Engage L2 Engineer	✓	✓	✓	✓
Alert BU Head	✓	✓	✓	✓
Raise Case with Cisco TAC	✓	✓	✓	✓
Alert General Manager	✓	✓	✓	✓

The Helpdesk is capable to assign and distinguish between multiple escalation criteria for different customers based on the signed SLA's

3.4.6.4 Call logging

- All incidents/problems are logged with the Helpdesk where a ticket is opened, necessary information gathered, Support contract validated, and a severity assigned to the ticket.
- An auto generated email is sent to the client and the concerned people by the system.
- Following this the Helpdesk assigns an engineer for the ticket

3.4.6.5 Case Tracking

- For each problem, a case number will be provided to HCT.
- HCT can request a detailed case report for any given case number
- MDS Call Center has a tracking and reporting facility that will provide following reports upon request:
 - Number of Logged Calls

- Logged Calls status
- Detailed Report for logged Call per month

Note: Additional reports can be customized upon request. These reports will be subject to both parties' approval.

3.4.6.6 Closure, Client Feedback & Satisfaction

- Once a ticket is reported closed by the assigned engineer, a call center agent calls the client to make sure that the issue has been resolved as per their satisfaction before closing the ticket.
- If there are any pending tasks or if the work has not been performed as per the required standards, the engineer is directed to take the necessary actions.

3.4.6.7 Periodic Reports & Review

Tracking reports can be provided to the customer to:

- Review services delivery since the last review
- Resolve any conflicts or concerns about service delivery.
- Re-evaluate services considering changes in business needs and resources.
- Discuss changes planned or in progress to improve service effectiveness.

3.4.7 HOW TO LOG A TICKET

MDS has a central help desk for logging support calls and providing Level-1 Support.

- Working hours are from **8:00 am to 5:00 pm**, Sunday to Thursday.
- Agents can be reached (8:00 am to 5:00 pm) through:
- Dialing the toll-free number **800-2333 OR 024046888**
- Sending e-mail to **helpdesk@mdsuae.ae** (OR)
- For 24x7 support calls MDS will provide with the Phone number of the Technical team lead and the engineer at the time of **contract signing**. These details will be provided in the SLA Agreement which will be signed after project award.

3.5 PROPOSED VALUE ADD

MDS offers implementation & consulting services designed to help your organization achieve success as quickly as possible through its expert teams. These services are aligned with our implementation methodology.

This section provides details about the services which will be provided to install, configure and provision the offered products. This section also lists any services or work which is not covered by MDS as part of the offered services.

- Cabling
- Annual Support Contract for three years

3.6 SOLUTION INTRODUCTION

This proposal details the Palo Alto Networks solution for the Higher Colleges of Technology (HCT) Request for Proposal (RFP) NE-26-0129, concerning a 3-year Secure Access Service Edge (SASE) platform. Our response is structured to align with the RFP, providing a comprehensive overview of our proposed solution, project methodology, and commercial terms.

3.6.1 TERMINOLOGIES

Palo Alto Networks confirms its understanding and adherence to all terminologies defined in Section 1.1 of the RFP. Our proposal is written with these definitions in mind to ensure clear and consistent communication.

3.6.2 GLOSSARY OF TERMS

Palo Alto Networks confirms its understanding of the terms “Request for Proposal (RFP),” “Bidder,” and “Proposal” as defined in Section 1.2 of the RFP. We will use these terms consistently throughout our response.

3.6.3 STRUCTURE OF THE DOCUMENT

This proposal is structured to directly address the sections outlined in the RFP, providing a clear and comprehensive response to HCT’s requirements. The document is organized as follows:

- **Section 1: Introduction:** Provides an overview of our understanding of HCT’s needs and our proposed solution.
- **Section 2: Evaluation and Guidelines:** Details Palo Alto Networks’ corporate profile, technical expertise, and relevant experience, aligning with HCT’s evaluation criteria.
- **Section 3: Technical Requirements:** A detailed response to the technical scope of work, addressing each requirement from sections 3.1 to 3.25 of the RFP.

- **Section 4: Appendices:** Includes the completed Technical Requirements Compliance matrix as requested in Appendix-B.
- **Section 5: Commercial Guidelines:** Presents the commercial offer structure as requested in Section 6 of the RFP.
- **Section 7: Technical Evaluation Criteria:** Summarizes how our proposal meets the specific evaluation criteria outlined by HCT.

3.6.4 ABOUT HCT

Palo Alto Networks acknowledges the esteemed position of the Higher Colleges of Technology (HCT) as the UAE's largest applied higher education institution. We understand HCT's strategic focus on technology-driven innovation, its mission to foster Emirati excellence, and its commitment to providing a secure, state-of-the-art learning environment for its 25,000+ students across 17 campuses. Our proposed SASE solution is designed to support HCT's strategic pillars—Talent Engine, Catalyst for Transformation, and Thriving Community—by providing a secure and agile foundation for digital education and collaboration.

3.6.5 BACKGROUND

We understand HCT's current network security landscape, which includes a fragmented approach with a legacy on-premises solution for BYOD users and a separate cloud-based solution for managed users. This introduces operational complexity, inconsistent policies, and visibility gaps. HCT's plan to implement a unified, fully cloud-based SASE platform to consolidate security, reduce costs, and prepare for a cloud-first strategy with local internet breakouts is a forward-thinking initiative that aligns perfectly with industry best practices and the core value proposition of Palo Alto Networks Prisma SASE.

3.6.6 OBJECTIVES AND OUTCOME

Palo Alto Networks' proposal is designed to meet all of HCT's stated objectives and deliver the desired outcomes. Our Prisma SASE solution will:

- **Unified Internet Security:** Consolidate HCT's legacy systems into a single, cloud-delivered platform, Prisma Access, for all internet traffic from both managed and BYOD users.
- **Enhanced Security:** Deliver superior security through our AI-powered threat prevention engine, which provides real-time, inline inspection to stop zero-day threats, phishing, and advanced malware.
- **BYOD and Managed User Parity:** Our unified agent (Prisma Access Agent) for managed devices and agentless solution (Prisma Browser) for BYOD ensure consistent policy enforcement and a secure experience for all users, regardless of device ownership.
- **AI-Powered Protection:** Leverage Precision AI™ for real-time threat detection, User and Entity Behavior Analytics (UEBA), automated incident response, and granular governance of Generative AI applications.
- **Cloud Optimization:** Provide secure, low-latency access to cloud applications through our global network of PoPs and intelligent traffic steering, fully supporting HCT's cloud migration.

- **Zero Trust Integration:** Seamlessly integrate with HCT's existing ecosystem (NGFW, NAC, IdP, EDR, SIEM) to establish a comprehensive ZTNA 2.0 security model.
- **Compliance:** Help HCT meet and exceed compliance requirements for FERPA, NIST CSF, ISO 27001, and UAE data protection regulations through robust security controls, data residency options, and detailed auditing.
- **Simplified Management:** Provide a single, intuitive management console, Strata Cloud Manager, for centralized policy management, reporting, and analytics.
- **Scalability:** Offer a massively scalable, cloud-native architecture that will effortlessly accommodate HCT's future growth in users, campuses, and cloud services.
- **Optional Zero Trust Private Access (ZTNA):** Our proposal includes a fully-costed option to replace legacy VPN with our ZTNA 2.0 solution, providing secure, identity-aware access to private applications for 200 designated users.

3.6.7 CONFIDENTIALITY

Palo Alto Networks fully agrees to the confidentiality terms outlined in Section 1.7. We will treat all information designated as "Confidential Information" by HCT with the highest degree of care, using it solely for the benefit of HCT and for the purposes of this engagement. We will not disclose, copy, or distribute any Confidential Information to any third party without HCT's explicit written consent.

3.6.8 INDEMNIFICATION

Palo Alto Networks agrees to the indemnification terms as stated in Section 1.8. We will defend, indemnify, and protect HCT from and against all lawsuits and costs, including reasonable legal fees, arising from any claim that our software infringes upon the intellectual property rights of any third party.

3.7 EVALUATION CRITERIA

This section details Palo Alto Networks' qualifications, experience, and proposed team structure, directly addressing the evaluation criteria outlined in the RFP. We are confident that our proven leadership in SASE, extensive experience with large-scale deployments, and world-class technical team make us the ideal partner for HCT.

3.7.1 EVALUATION CRITERIA

Our proposal is structured to provide a clear and comprehensive response to each of HCT's evaluation criteria.

3.7.1.1 Companies' Profile and General Information Scoring

- **Similar Success Stories and Client References:** Palo Alto Networks has a proven track record of successful SASE deployments for numerous government and educational institutions worldwide. These organizations have modernized their security, enabled secure remote work, and reduced operational complexity by adopting Prisma SASE. Success stories from entities such as the Government of South Australia Department for Education, Amplify, and Monroe County demonstrate our expertise in securing complex, distributed environments similar to HCT's. We are pleased to provide detailed client references upon request.
- **Company Profile, Technical Staff, and Certifications:** Palo Alto Networks is the global leader in AI-powered cybersecurity. With thousands of highly qualified and certified engineers, including systems engineers, R&D specialists, and technical support staff, we possess unparalleled technical depth. Our support staff hold industry-recognized certifications, including our own rigorous PCNSE (Palo Alto Networks Certified Network Security Engineer) certification, ensuring expert-level assistance. We have a strong local presence in the UAE and the broader GCC region to provide effective project execution and support.
- **Partnership Level with Principal Vendors:** As the principal technology provider and developer of the entire proposed Prisma SASE solution, Palo Alto Networks offers a fully integrated, single-vendor platform. This eliminates the multi-vendor complexity, integration challenges, and security gaps often associated with piecemeal solutions. HCT will benefit from a seamless architecture, unified support, and a single, accountable partner.
- **Service Desk/Helpdesk System and Support Team:** Palo Alto Networks provides world-class, 24/7/365 global support through our Customer Support Portal (CSP) and tiered Premium Success plans. Our dedicated support team includes Technical Assistance Center (TAC) engineers and Customer Success teams who provide personalized guidance, proactive support, and rapid incident resolution. We offer industry-leading SLAs for response times, ensuring HCT receives timely and expert assistance for any issue.
- **Financial Stability and Business Viability:** Palo Alto Networks is a publicly traded company (NASDAQ: PANW) with a strong and stable financial profile. As a consistent market leader with over 70,000 customers, including 95 of the Fortune 100, we have a proven track record of profitability and strong cash flow. Our financial strength ensures our long-term viability and our ability to support HCT's quarterly payment structure over the 3-year contract period and beyond.

3.7.1.2 Technical Team

- **Dedicated Onsite/Offsite Resources:** Palo Alto Networks, in partnership with a certified local implementation partner, will provide a dedicated team of resources, including a Resident Engineer. Our model ensures 100% backfill in the absence of the primary Resident Engineer, guaranteeing continuous, expert support for HCT.
- **24/7/365 Availability:** Our global support infrastructure and follow-the-sun model ensure that expert resources are available 24/7/365 to support all HCT locations.
- **Detailed Resumes:** We will provide detailed resumes for all key personnel assigned to this project upon request, showcasing their extensive experience and relevant certifications.
- **Team Certification Level:** Our proposed team members hold the highest levels of industry and Palo Alto Networks certifications, including PCNSE, CISSP, and cloud-specific credentials, ensuring they are experts in the proposed technologies.

3.7.1.3 Technical Criteria and Scope Compliance

- **Compliance with Technical Requirements (Section 3.1-3.10):** This proposal provides a detailed, point-by-point response demonstrating how our Prisma SASE solution fully complies with all technical requirements outlined in Sections 3.1 through 3.10 of the RFP.
- **Compliance with Warranty and Support (Section 3.11):** We confirm our ability to meet the managed support services requirements through a combination of our world-class Premium Support and a certified local partner.
- **Compliance with Security Requirements (Section 3.19):** Our platform is designed to help organizations meet stringent compliance mandates, and this proposal details our capabilities regarding FERPA, NIST CSF, ISO 27001, PCI-DSS, and UAE data protection regulations.

3.8 TECHNICAL SCOPE OF WORK

This section provides a detailed response to the technical requirements outlined in the RFP.

3.8.1 GENERAL SCOPE OF WORK

Palo Alto Networks' Prisma SASE is a fully cloud-delivered SASE architecture that fully complies with the general scope of work outlined in RFP Section 3.1.

- **1 Fully Cloud-Delivered SASE Architecture:** Prisma SASE is a fully cloud-native platform that converges networking and security into a single, unified service. It provides centralized management via Strata Cloud Manager, built-in redundancy through its hyperscale architecture, and AI-powered threat prevention for both managed and BYOD users.
- **2 Converged SASE Solution:** Prisma Access, the core of our SASE solution, unifies Secure Web Gateway (SWG), Firewall-as-a-Service (FWaaS), Remote Browser Isolation (RBI) via our Secure Enterprise Browser (Prisma Browser), and DNS Security into a single, integrated platform.
- **3 Detailed Deployment Plan:** We will provide a comprehensive, phased deployment plan to migrate all HCT campuses, the Primary DC, and the DR site to Prisma SASE. This plan includes seamless integration with local internet breakout links at each campus, ensuring a smooth transition with minimal disruption.
- **4 Agent-Based and Agentless Deployment:** Our solution fully supports both deployment models. The lightweight **Prisma Access Agent** provides comprehensive security for managed (institution-owned) endpoints. For BYOD (personally-owned) endpoints, our agentless **Prisma Browser** and other clientless access methods provide secure access with differentiated security policies without requiring a full agent installation.
- **5 AI-Powered Security Capabilities:** Our platform is powered by Precision AI™, delivering industry-leading real-time threat detection, zero-day malware prevention, and advanced phishing detection. Our User and Entity Behavior Analytics (UEBA) capabilities detect anomalous behavior, while our AI Access Security provides granular controls for Generative AI applications.

Figure 5: This conceptual diagram illustrates the flow of traffic through Prisma SASE, where Precision AI™ performs real-time threat detection and prevention, extending Zero Trust security and granular controls to Generative AI applications.

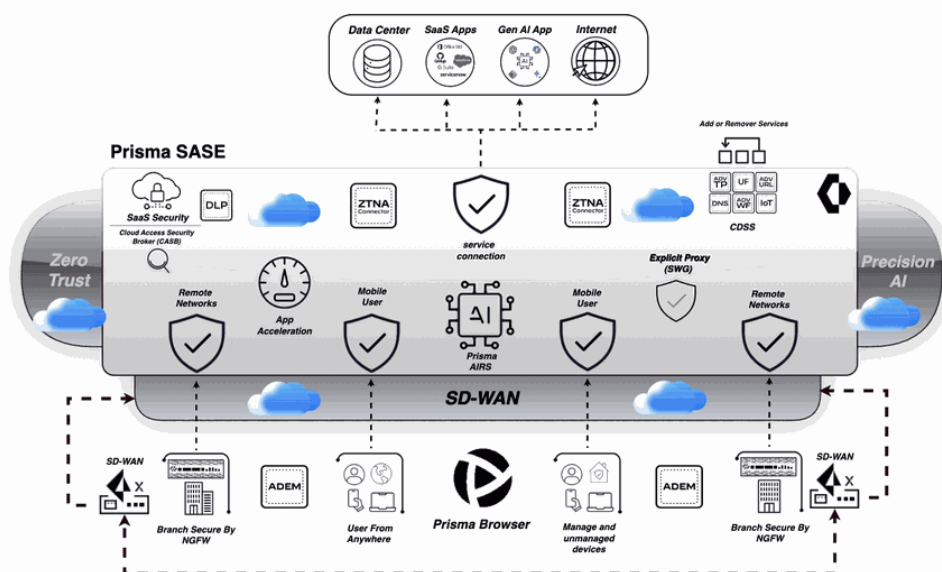


Figure 5: This conceptual diagram illustrates the flow of traffic through Prisma SASE, where Precision AI™ performs real-time threat detection and prevention, extending Zero Trust security and granular controls to Generative AI applications.

- 6 Migration and Consolidation Plan:** We will execute a structured migration plan to consolidate HCT's legacy on-premises SASE for BYOD and the existing cloud solution for managed users onto the unified Prisma SASE platform, eliminating security silos and operational complexity.
- 7 Inline SSL/TLS Inspection:** Prisma Access provides robust inline SSL/TLS inspection for all internet-bound traffic. It supports the use of HCT's local CA-signed certificate, re-encrypts traffic using TLS 1.2 and 1.3, and enables advanced threat prevention, content filtering, and application control on de-crypted traffic.
- 8 Real-Time Monitoring and DEM:** Our integrated Autonomous Digital Experience Management (ADEM) provides real-time, end-to-end visibility into user experience, network performance, and security incidents, enabling proactive troubleshooting and rapid root-cause analysis.
- 9 Training and Documentation:** Palo Alto Networks and our certified partners will provide comprehensive training for HCT's IT staff and deliver detailed documentation, including architectural diagrams, as-built guides, and runbooks, to ensure a successful handover and operational readiness.
- 10 Secure Internet Breakouts:** Prisma Access is designed to secure local internet breakouts at the campus level. We support IPsec tunnel-based traffic steering from campus edge devices (NGFW or SD-WAN appliances) to the nearest Prisma Access PoP for low-latency, secure internet access.
- 11 Agentless BYOD Security:** Our agentless BYOD security is delivered through multiple mechanisms, including DNS-layer security, PAC file distribution for explicit proxy, certificate-based authentication, and Remote Browser Isolation (RBI) integrated within our Prisma Browser.
- 12 SaaS Application Visibility and Control:** Our integrated Next-Gen CASB provides granular visibility and control over thousands of SaaS applications, enabling shadow IT discovery and tenant-level restrictions for multi-tenant platforms to prevent data exfiltration.

- **13 Granular Controls for Generative AI:** Our AI Access Security solution offers industry-leading controls for GenAI usage, including automated discovery of GenAI applications, per-application policies (e.g., block, read-only, full access), and detailed analytics on AI usage and data exposure risks.
- **14 Integration with Existing Infrastructure:** Prisma SASE is designed for seamless integration with HCT's existing ecosystem, including Identity Providers (IdP), Network Access Control (NAC), Next-Generation Firewalls (NGFW), Endpoint Detection and Response (EDR/XDR), SIEM/SOAR, and Unified Endpoint Management (UEM/MDM) solutions.
- **15 Simplified Operations:** Strata Cloud Manager provides a single, centralized cloud management console for all SASE functions. Its intuitive interface and AI-powered operational assistance simplify policy management and integrate seamlessly with HCT's managed SOC for unified visibility and response.
- **16 Compliance Support:** Our solution is architected to help HCT meet and demonstrate compliance with key regulations, including FERPA, NIST CSF, ISO 27001, PCI-DSS, and UAE data protection laws, with features for data residency, robust encryption, and detailed audit logging.
- **17 -3.1.20 Optional ZTNA Capabilities:** Our optional ZTNA 2.0 offering provides secure, identity-aware, application-level access for 200 users to private applications without exposing the corporate network. It is governed by the same centralized SASE management console, policy engine, and logging infrastructure, ensuring a truly unified security model.

3.8.2 SECURE WEB GATEWAY (SWG) REQUIREMENTS

Palo Alto Networks Prisma Access delivers a comprehensive, cloud-native Secure Web Gateway (SWG) that meets all of HCT's requirements for securing HTTP/HTTPS internet-bound traffic for both managed and BYOD users.

- **Inline SWG Capabilities:** Prisma Access provides best-in-class inline SWG capabilities, inspecting all web traffic for both managed and BYOD users to enforce security policies and prevent threats in real-time.
- **Full Inline SSL/TLS Decryption:** Our solution performs full inline SSL/TLS decryption using HCT's local CA-signed certificate. It supports re-encryption of TLS 1.2 and TLS 1.3 traffic and includes configurable bypass policies for sensitive categories like finance and healthcare, as well as for applications that use certificate pinning.
- **Granular URL Categorization:** Our Advanced URL Filtering service offers over 70 granular content categories, far exceeding the minimum requirement of 85. It supports flexible policy actions including allow, block, caution (with user coaching), and isolate (via Remote Browser Isolation). Our engine uses real-time AI/ML to continuously discover and re-categorize newly observed URLs, ensuring protection against zero-day web threats.
- **Inline File Scanning and Sandboxing:** We provide multi-layered inline file scanning. This includes multi-engine antivirus, heuristic analysis, and our industry-leading **Advanced WildFire** cloud-based sandbox for detonation and analysis of unknown files (including PE, Office, PDF, and archive formats). This protection is applied consistently to both managed and BYOD traffic streams.
- **File Type and MIME Type Control:** Our platform allows granular control over file uploads and downloads based on true file type, not just extension. MIME type inspection is used to identify and control web content and enforce security policies.

- **Application-Aware Bandwidth Management:** Prisma Access includes robust Quality of Service (QoS) capabilities for application-aware bandwidth management. HCT can prioritize critical applications like Learning Management Systems (LMS) and e-learning traffic while throttling or limiting non-essential categories like streaming media, especially during peak academic hours.
- **Customizable Block and Coaching Pages:** We provide fully customizable block and coaching pages. HCT can incorporate its institutional branding, logo, and specific messaging. These pages can reference acceptable use policies and provide clear notifications to users about policy violations.
- **DNS Security:** Our Advanced DNS Security service inspects all DNS traffic, including DNS-over-HTTPS (DoH) and DNS-over-TLS (DoT). It uses AI/ML to detect and block DNS tunneling, domains generated by algorithms (DGA), newly registered domains, and known malicious domains in real-time.
- **Advanced Browser Threat Protection:** Our unique **Prisma Browser** provides AI-powered inspection of fully rendered web content (post-DOM construction). This detects and stops evasive threats that other proxies miss, such as JavaScript-based credential theft, post-load payload assembly, and sophisticated browser exploitation techniques.
- **Encrypted Traffic Analysis:** Our platform leverages AI/ML to analyze encrypted traffic metadata without requiring decryption in all cases. By inspecting metadata such as JA3/JA4 fingerprints, certificate attributes, and traffic patterns, we can detect malicious communications and C2 channels hidden within encrypted flows.
- **TLS Inspection Exemption:** Prisma Access supports highly granular TLS inspection exemption policies. Exemptions can be configured based on site reputation databases, web categories (e.g., financial, healthcare), specific source/destination addresses, or user/group identity.
- **Proxy Avoidance Prevention:** Our solution includes multiple mechanisms to prevent proxy avoidance. This includes blocking the “proxy-avoidance-and-anonymizers” URL category, rate-limiting suspicious URL patterns, blocking malicious redirects, and using App-ID to identify and block known proxy applications and behaviors.

3.8.3 FIREWALL-AS-A-SERVICE (FWaaS) REQUIREMENTS

Palo Alto Networks Prisma Access delivers a true cloud-native Firewall-as-a-Service (FWaaS) that provides NGFW-equivalent capabilities for all of HCT’s internet-bound traffic, meeting all specified requirements.

- **Cloud-Delivered NGFW Capabilities:** Prisma Access is a fully cloud-delivered service that provides the complete security stack of a Palo Alto Networks Next-Generation Firewall (NGFW), ensuring consistent and comprehensive protection for all users and locations without the need for on-premises hardware.
- **Layer 7 Application Identification and Control:** Our industry-leading App-ID technology identifies and controls over 5,000 applications and protocols, regardless of port, encryption, or evasive tactics. It provides granular control over sub-applications, allowing HCT to create policies that, for example, permit Teams chat while blocking Teams file transfers.
- **Inline Intrusion Prevention System (IPS):** Our Advanced Threat Prevention service provides a comprehensive inline IPS. It uses a combination of signature-based and anomaly-based detection, with automatic signature updates from our global threat intelligence network. It provides robust coverage for the OWASP Top 10, known CVE-based exploits, and protocol anomalies. Our threat database contains far more than the 13,000 required signatures, protecting against a vast landscape of threats.

- **IP, FQDN, Device, and User/Group-Based Policies:** Prisma Access supports highly granular firewall policies. Rules can be created based on source/destination IP addresses, Fully Qualified Domain Names (FQDNs), device posture (via HIP checks), and user/group identity (via integration with HCT's directory services).
- **IP Reputation-Based Deny Policies:** Our platform supports deny policies based on IP Reputation for a wide range of malicious categories, including Botnet-C&C, Phishing, Proxy servers, Spam sources, and TOR nodes. This intelligence is delivered via continuously updated, vendor-provided signature and threat intelligence feeds.
- **Geo-IP Filtering:** Prisma Access provides robust Geo-IP filtering capabilities, allowing HCT to allow or block traffic based on source and destination country. Configurable exception lists can be created to tailor policies to specific security needs.
- **Protocol Enforcement:** Our solution enforces strict protocol validation. This includes deep HTTP protocol validation to prevent evasion, SSH version control to block weak or vulnerable versions, and the ability to block non-standard protocol usage over standard ports using our App-ID technology.
- **HTTP/3 and QUIC Detection:** Prisma Access can detect traffic using the HTTP/3 (QUIC) protocol. To ensure full inspection, our platform can be configured to block QUIC traffic, forcing browsers to fall back to HTTP/2 or HTTP/1.1 with TLS 1.2/1.3, which can then be fully decrypted and inspected.
- **DNS-Based Filtering and Botnet Blocking:** Our Advanced DNS Security provides granular DNS-based filtering for security risk categories such as Dynamic DNS, Malicious Websites, Newly Observed/Registered Domains, Phishing, and Spam domains. Furthermore, we provide robust protection against botnets through both IP-based blocking (using a global IP reputation database) and domain-based blocking to disrupt Fast Flux botnet domains.

3.8.4 REMOTE BROWSER ISOLATION (RBI)/ ENTERPRISE BROWSER REQUIREMENTS

Palo Alto Networks meets and exceeds the requirements for RBI through our innovative **Prisma Browser**, a secure enterprise browser that provides superior security and user experience compared to traditional pixel-based RBI solutions.

- **Pixel-Based Isolation:** While we offer traditional pixel-based RBI, our recommended and superior approach is the **Prisma Browser**. It executes web content locally within a hardened, secure browser environment, but with security inspection and policy enforcement happening in-browser before content is rendered. This provides the security of isolation without the latency and user experience degradation inherent in streaming pixels from a remote container. It effectively prevents active web content (JavaScript, etc.) from exploiting the underlying OS or other applications.
- **Policy-Driven Isolation:** Prisma Browser supports highly granular, policy-driven security. Policies can be based on URL category (e.g., isolate all uncategorized or newly registered domains), user risk score (dynamically applying stricter controls for high-risk users), or device posture (enforcing different rules for managed vs. BYOD devices).
- **Read-Only Mode for BYOD:** Prisma Browser provides a configurable read-only mode, ideal for BYOD users accessing sensitive web applications. This mode can be enabled via policy to prevent copy, paste, download, upload, and printing from the browser session, effectively preventing data exfiltration.

- **Transparent User Experience:** Prisma Browser is built on Chromium, providing users with the familiar, high-performance experience they expect. By performing inspection locally and avoiding pixel streaming, it eliminates the latency, rendering issues, and broken web pages common with traditional RBI. It fully supports modern web applications, rich media, single-page applications, and collaboration tools, ensuring a seamless and productive user experience.
- **Clientless ZTNA via Browser-Based RBI:** Prisma Browser provides a truly clientless ZTNA experience for BYOD users accessing institutional web applications. It enforces Zero Trust principles directly within the browser, verifying user and device posture before granting access, without requiring any agent installation on the endpoint.

3.8.5 AI-POWERED SECURITY REQUIREMENTS

Palo Alto Networks is the leader in applying AI and machine learning to cybersecurity. Our Prisma SASE platform is infused with AI-powered security capabilities to provide unparalleled protection.

3.8.5.1 AI-Driven Threat Detection and Prevention

- **AI-Based Phishing Detection:** Our Advanced URL Filtering service uses multiple AI and ML engines for real-time analysis of web page content, URL patterns, brand impersonation indicators, and page behavior. This allows us to detect and block sophisticated, zero-day phishing pages at time of access, even those that dynamically assemble malicious content after the initial page load.
- **Zero-Day Malware Detection:** Our Advanced WildFire service uses a combination of ML-based static and dynamic analysis of files in the inline inspection path. This multi-faceted approach allows us to identify previously unseen malware with high efficacy, without relying on traditional signature databases.
- **AI-Powered DNS Threat Detection:** Our Advanced DNS Security service employs multiple machine learning models to analyze DNS query patterns in real time. This enables the detection of domain generation algorithms (DGA), DNS tunneling for data exfiltration, and suspicious newly observed domains, blocking threats at the earliest point in the attack lifecycle.
- **Advanced Browser Threat Protection:** Prisma Browser provides AI-powered inspection of fully rendered web content (post-DOM construction). This unique capability allows us to detect and neutralize evasive threats that traditional proxies miss, such as post-load malware assembly and sophisticated browser exploitation techniques.
- **Encrypted Traffic Analysis:** We leverage AI/ML to analyze encrypted traffic metadata (such as JA3/JA4 fingerprints, certificate attributes, and traffic patterns) to detect malicious communications, like C2 channels, without needing to decrypt the traffic in all cases, balancing security and privacy.

3.8.5.2 AI-Based User and Entity Behavior Analytics (UEBA)

- **Behavioral Baseline:** Our UEBA engine, integrated within Prisma Access, continuously learns and establishes a dynamic baseline of normal internet usage patterns for each user and group. This includes typical access times, data volumes, application usage, and geographic locations.
- **Anomaly Detection:** The platform provides real-time flagging of deviations from these established baselines. This includes detecting unusual data upload volumes, access to atypical application categories,

logins from anomalous locations, or the use of evasion tools like anonymizers and personal VPN services.

- **Risk Scoring:** We generate dynamic per-user and per-device risk scores that adapt based on observed behavior. These risk scores are then available as context for policy enforcement, allowing HCT to, for example, subject high-risk users to stricter inspection, enforce step-up authentication, or automatically apply browser isolation.

3.8.5.3 AI-Powered Operational Intelligence

- **AI-Assisted Policy Recommendations:** Our Strata Cloud Manager platform analyzes traffic patterns and threat data to provide AI-assisted recommendations for policy optimization, helping HCT to tighten security posture and improve efficiency.
- **AI Ops for Troubleshooting:** Our Autonomous Digital Experience Management (ADEM) provides AI-driven root cause analysis that correlates user experience degradation with underlying issues, whether they are network path problems, PoP performance degradation, or policy conflicts, dramatically accelerating incident resolution.
- **Predictive Capacity Management:** ADEM uses AI to forecast traffic volumes and PoP utilization. This predictive capability allows HCT to proactively plan for capacity needs and prevent performance degradation during peak periods like the start of an academic term or examination weeks.

3.8.5.4 Generative AI Application Controls

Figure 6: AI Access Security features for Generative AI, including automated application discovery, granular policy controls, and usage analytics, supported by Enterprise DLP and real-time user coaching to manage data exposure risks.

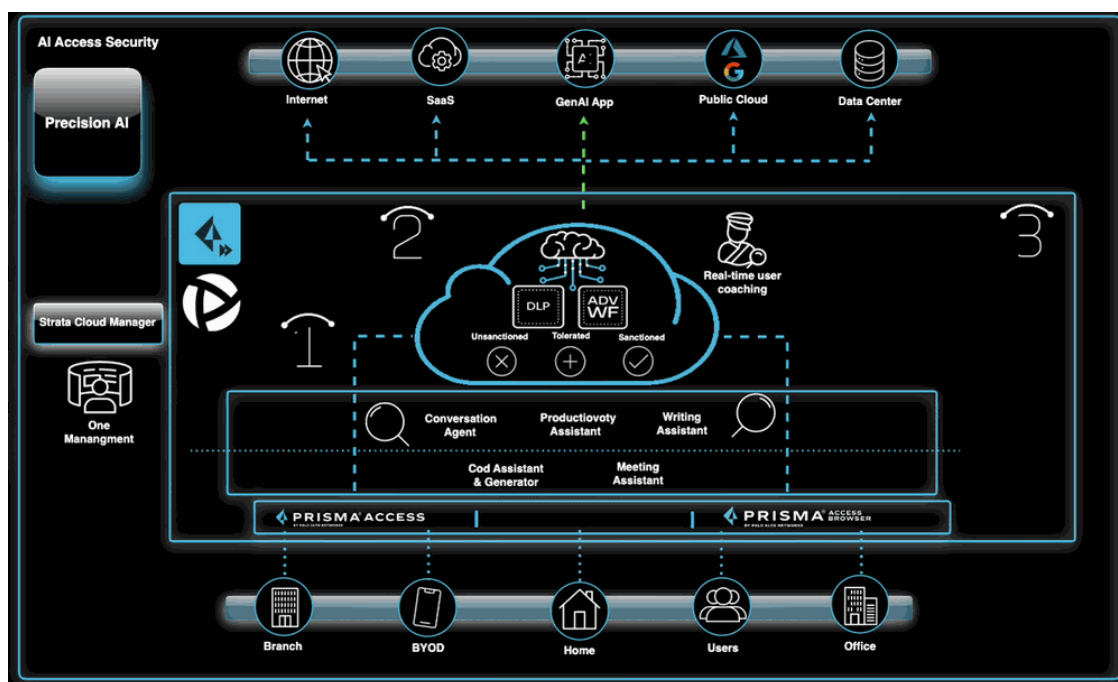


Figure 6: AI Access Security features for Generative AI, including automated application discovery, granular policy controls, and usage analytics, supported by Enterprise DLP and real-time user coaching to manage data exposure risks.

- **GenAI Application Discovery:** Our AI Access Security solution provides automated identification and categorization of all Generative AI applications (e.g., ChatGPT, Gemini, Copilot, Midjourney) accessed through the platform, giving HCT full visibility into GenAI usage.
- **Granular GenAI Policy Controls:** We offer granular, per-application controls that allow HCT to define differentiated policies. For example, you can allow ChatGPT for staff but restrict it to read-only for students, block file uploads to unsanctioned AI tools, and grant full access to institutionally-approved AI tools.
- **AI Usage Analytics:** Our platform includes dedicated dashboards and reports showing GenAI application adoption trends, data exposure risks, and policy violations across different user groups, providing actionable intelligence for governance.
- **Agentic AI Governance:** We provide visibility into agent-to-tool communications, intent-aware inspection of interactions and tool requests, and granular policy controls to govern agent connectivity and prevent misuse.

3.8.6 BYOD-SPECIFIC REQUIREMENTS

Palo Alto Networks provides a multi-faceted approach to securing BYOD users, offering flexibility and robust security without requiring a full endpoint agent.

- **PAC File / Proxy Auto-Configuration:** We fully support browser-level traffic steering via PAC files. HCT can use DHCP/WPAD or manual configuration to distribute a PAC file that directs BYOD browser traffic to our Prisma Access cloud for security inspection.

- **DNS Redirection:** Our solution enables DNS-layer security by allowing HCT to configure campus DHCP servers to point BYOD clients to the Prisma Access DNS resolvers. This provides baseline protection and policy enforcement at the DNS layer without any agent or proxy configuration on the endpoint.
- **IPsec Tunnel from Network Edge:** We support steering all internet traffic from designated BYOD VLANs to the nearest SASE PoP via an IPsec tunnel established from a campus edge device (e.g., firewall or router). This method transparently captures all traffic from BYOD devices on that network segment for full inspection.
- **Certificate-Based Authentication:** Our platform supports certificate-based authentication for BYOD users. Through an institutional onboarding portal using SCEP/EST, certificates can be distributed to BYOD devices, providing strong identity context for policy enforcement without requiring a full agent.
- **Device Type Differentiation:** Our solution can differentiate between managed and BYOD devices using several methods, including the presence of a management agent, device certificates, or device posture attributes collected by Prisma Browser. This allows for the application of differentiated security policies.
- **Privacy-Preserving Inspection:** For BYOD users, we offer privacy-preserving inspection through selective SSL inspection. Policies can be configured to bypass decryption for sensitive categories like personal banking and healthcare. Our Prisma Browser also provides clear user notifications about inspection status, ensuring transparency.
- **Session-Based Controls:** We provide robust session-based controls for BYOD users, including configurable maximum session durations and re-authentication intervals to ensure continuous identity verification and mitigate risks from unattended sessions.
- **Guest User Self-Registration:** Our solution integrates with captive portal workflows for guest user self-registration. This includes presenting terms-of-use for acceptance, providing time-limited access, and applying configurable bandwidth quotas to manage guest network usage.

3.8.7 MANAGED USER-SPECIFIC REQUIREMENTS

For HCT's managed devices, Palo Alto Networks provides a powerful, unified endpoint agent that delivers comprehensive SASE functionality.

Figure 7: The Prisma Access Agent on a managed endpoint provides always-on, comprehensive SASE functionality, enabling remote users to establish secure TLS/SSL and IPsec connections to Prisma Access for safe access to private applications and the internet.

Palo Alto Networks Prisma Access Agent
Common Design

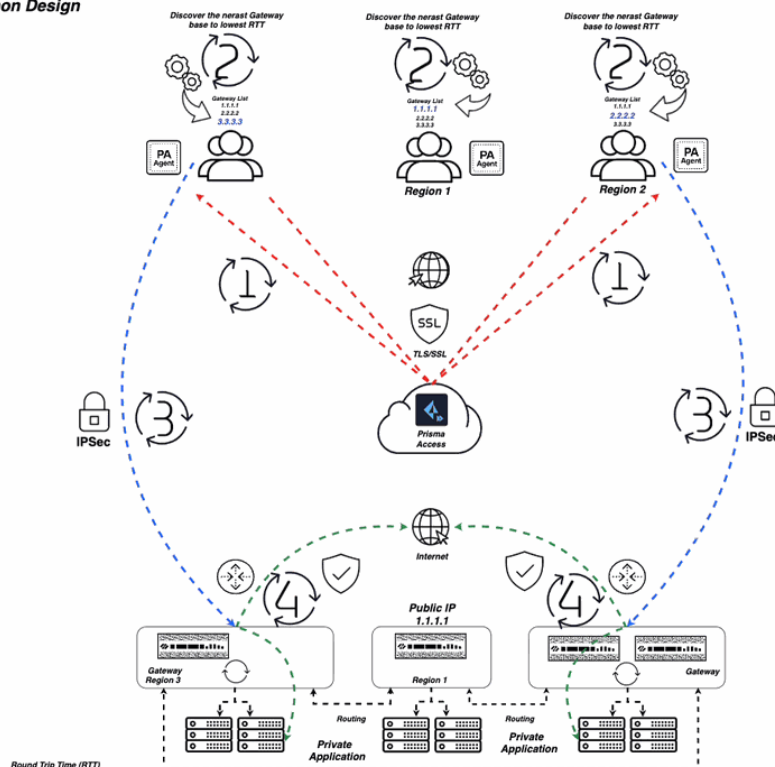


Figure 7: The Prisma Access Agent on a managed endpoint provides always-on, comprehensive SASE functionality, enabling remote users to establish secure TLS/SSL and IPsec connections to Prisma Access for safe access to private applications and the internet.

- Unified Endpoint Agent:** Our **Prisma Access Agent** is a single, lightweight agent that handles all SASE functions, including SWG traffic steering, DNS security, certificate management, and device posture reporting. The agent supports Windows 10/11, macOS (latest and N-1), iOS, and Android.
- Always-On Connectivity:** The Prisma Access Agent enforces an always-on, secure tunnel to the nearest SASE PoP, with no ability for the user to disable it. This ensures that all traffic is continuously protected. The agent's behavior is configurable to either fail-open (allow traffic) or fail-close (block traffic) if the SASE PoP is unreachable.
- Device Posture Assessment:** The agent performs continuous endpoint posture checks, verifying OS patch level, endpoint protection (EPP) status, disk encryption status, firewall status, domain join status, and certificate validity. These posture results feed directly into the policy engine for dynamic access decisions.
- Per-App Traffic Steering:** The Prisma Access Agent supports granular, per-application traffic steering. HCT can define policies to selectively route specific application traffic through the SASE tunnel while allowing configurable split tunneling for trusted local services, such as local printer traffic or access to campus LAN resources.
- Agent Tamper Protection:** The agent includes robust anti-tamper mechanisms that prevent users, even those with local administrator credentials, from uninstalling, disabling, or circumventing the agent.

- **Silent Deployment:** The Prisma Access Agent supports silent deployment and updates via enterprise endpoint management tools like Microsoft SCCM, Intune, and Jamf, allowing for seamless, large-scale rollouts without user intervention.
- **(Optional ZTNA) Unified Agent for Internet and Private Access:** For the optional ZTNA scope, the same unified Prisma Access Agent will handle both internet traffic steering and ZTNA private application tunneling. No separate ZTNA agent is required. The agent will seamlessly route traffic to private applications through the SASE cloud ZTNA broker while simultaneously routing internet traffic through the SWG/FWaaS inspection path.
- **(Optional ZTNA) Split-Tunnel Intelligence:** The agent supports advanced split-tunnel intelligence that can distinguish between traffic destined for published private applications (routed via the ZTNA broker), general internet-bound traffic (routed via SWG/FWaaS), and local campus LAN traffic (routed directly). The traffic steering table is dynamically updated based on the applications published in the user's ZTNA policy.

3.8.8 PLATFORM INFRASTRUCTURE AND PERFORMANCE REQUIREMENTS

Palo Alto Networks' Prisma SASE platform is built on a massively-scaled, high-performance global network designed to deliver exceptional user experience and reliability.

- **Global PoP Footprint:** We operate a globally distributed network of Points of Presence (PoPs) across all major cloud providers. We have a strong presence in the UAE and the broader GCC region, ensuring we can meet the sub-30ms latency requirement from all HCT campus locations.
- **PoP Redundancy:** Our architecture ensures automatic and seamless failover between PoPs with no user-perceptible disruption. Each PoP is multi-homed with carrier-diverse internet connectivity for maximum resilience.
- **Capacity and Scalability:** Our cloud-native platform is horizontally scalable and designed to support the total concurrent user count of HCT (29,000+ users, 45,000 concurrent devices) with no performance degradation.
- **Data Residency:** We provide HCT with the ability to select the geographic region for storing logs and configuration data, ensuring compliance with institutional and regulatory data residency requirements.
- **Uptime SLA:** We offer an industry-leading, financially-backed service level agreement (SLA) of 99.999% platform availability.
- **Latency SLA:** We commit to a maximum additional latency of less than 25ms for SWG/FWaaS inspection and less than 50ms for full SSL inspection plus sandboxing.
- **Throughput SLA:** Our platform will support a minimum per-user throughput of 100 Mbps for managed users and 50 Mbps for BYOD users through the full inspection stack.
- **Digital Experience Monitoring (DEM):** Our integrated Autonomous DEM (ADEM) provides hop-by-hop visibility from the endpoint to the application, synthetic monitoring for critical SaaS applications, AI-powered root cause analysis, and user experience scoring to proactively identify and resolve performance issues.

- **(Optional ZTNA) ZTNA Session Establishment:** For the optional ZTNA scope, we guarantee session establishment times of less than 3 seconds for web applications and less than 5 seconds for thick client/RDP/SSH applications.
- **(Optional ZTNA) ZTNA Throughput:** We will support a minimum per-user throughput of 50 Mbps for RDP sessions and 100 Mbps for file transfer operations through the ZTNA inspection path.
- **(Optional ZTNA) ZTNA Latency:** The additional latency introduced by our ZTNA brokering path will not exceed 30ms for UAE-based users.

3.8.9 INTEGRATION REQUIREMENTS FOR ZERO TRUST ECOSYSTEM

Prisma SASE is designed with an open, API-first architecture to seamlessly integrate with HCT's existing and planned infrastructure, creating a cohesive Zero Trust ecosystem.

Figure 8: Prisma SASE acts as the central enforcement point for a cohesive Zero Trust ecosystem, delivering secure ZTNA to all applications with integrated Enterprise DLP and AI-driven security, and leveraging HCT's Identity Provider for robust authentication.

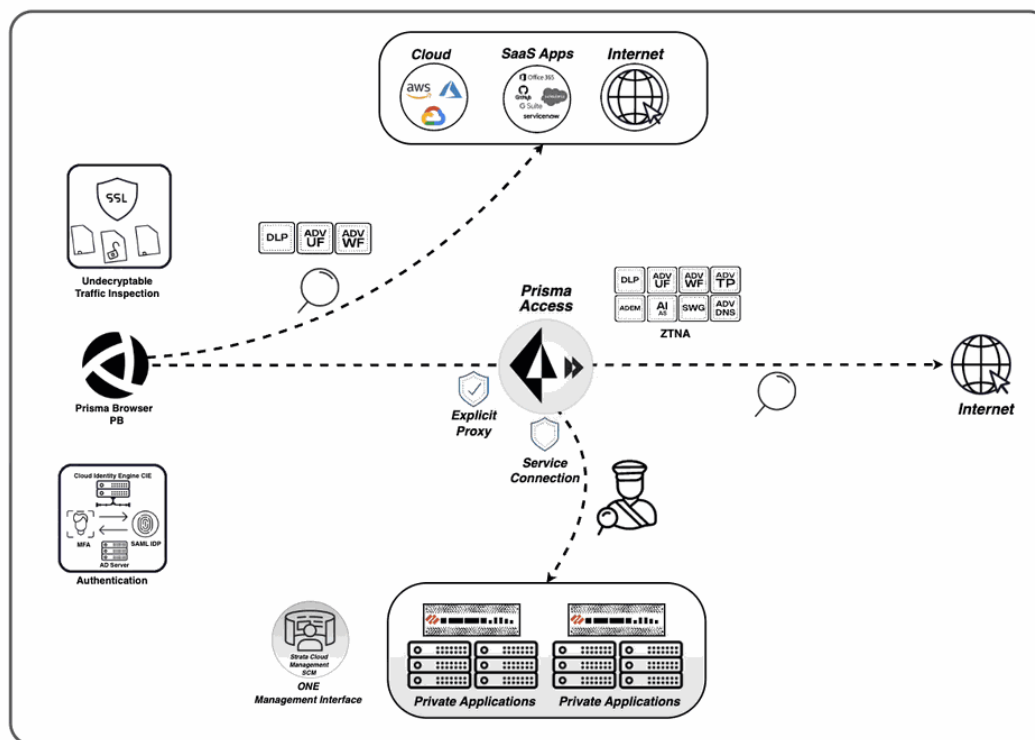


Figure 8: Prisma SASE acts as the central enforcement point for a cohesive Zero Trust ecosystem, delivering secure ZTNA to all applications with integrated Enterprise DLP and AI-driven security, and leveraging HCT's Identity Provider for robust authentication.

- **Identity Provider (IdP) + MFA:** We support SAML 2.0 and OIDC for integration with IdPs like Azure AD/Entra ID and HCT's in-house F5 SAML solution. We also support SCIM for automated user/group provisioning and can federate with on-premises AD. MFA can be enforced at the SASE layer for every access request. We also support EduRoam federation.

- **Network Access Control (NAC):** We integrate with NAC solutions via RADIUS CoA, REST API, or syslog to receive device posture and classification signals, allowing the SASE platform to differentiate policy for compliant vs. non-compliant devices.
- **Next-Generation Firewall (NGFW):** Our solution supports bi-directional log sharing (syslog/CEF/LEEF), shared threat intelligence via STIX/TAXII, and the ability to receive IPsec tunnels from HCT's existing NGFWs for on-premises traffic steering to the SASE PoP.
- **Endpoint Detection and Response (EDR/XDR):** We provide API-based integration to receive endpoint risk signals from EDR/XDR solutions. When an endpoint is flagged as compromised, our platform can dynamically adjust its access policy, such as restricting it to isolation-only browsing or blocking access entirely.
- **SIEM / SOAR:** We support real-time log export via syslog (CEF), API, or native connectors to all major enterprise SIEM platforms, including Splunk, Microsoft Sentinel, QRadar, and Elastic. Our platform also provides robust SOAR integration for automated response workflows.
- **Unified Endpoint Management (UEM/MDM):** Our solution can query device compliance status (e.g., encryption, OS version, patch level) from leading UEM/MDM platforms like Intune, Jamf, and VMware Workspace ONE, using this data as an input for SASE policy decisions.
- **Threat Intelligence Platform (TIP):** We can ingest external threat intelligence feeds (STIX/TAXII 2.1) and institutional Indicators of Compromise (IoCs) for custom blocklist enforcement.
- **ITSM / Ticketing:** We provide API integration with ITSM platforms like ServiceNow and Ivanti for automated ticket creation on security alerts and policy violations.
- **Microsoft Data Loss Prevention (DLP):** We can integrate with Microsoft DLP to enforce consistent data protection policies across web, cloud, and private application traffic.
- **Microsoft Defender for Cloud Apps (CASB):** Our solution integrates with Microsoft's CASB for enhanced cloud application visibility, SaaS security, and risk-based access control.
- **Single Sign-On:** We support deep integration with Windows AD, Microsoft Exchange, and Terminal Server Agents, as well as RADIUS Accounting and captive portal authentication for comprehensive SSO across wireless and wired users.
- **(Optional ZTNA) Privileged Access Management (PAM) Integration:** If HCT deploys a PAM solution, our ZTNA platform supports integration via API or SAML to enforce PAM-brokered sessions for RDP/SSH access, with session recording logs exportable to the PAM platform.
- **(Optional ZTNA) DNS-Based Application Routing:** Our ZTNA platform supports private DNS resolution, integrating with HCT's internal DNS servers to allow access to published applications via their internal FQDNs.
- **(Optional ZTNA) Service Ticket / Approval Workflow Integration:** Our ZTNA platform integrates with ITSM systems like Ivanti for access request workflows, enabling users to request time-limited ZTNA access to applications through an automated approval and provisioning/de-provisioning process.

3.8.10 CENTRALIZED MANAGEMENT AND REPORTING REQUIREMENTS

Palo Alto Networks Prisma SASE is managed through **Strata Cloud Manager**, a single, unified cloud-based console that meets all of HCT's management and reporting requirements.

- **Single Management Console:** All security and policy functions—including SWG, FWaaS, ZTNA, RBI, DNS Security, and DEM—are configured and monitored from a single, intuitive cloud portal. There are no separate consoles for different features.
- **Role-Based Access Control (RBAC):** We provide granular RBAC for administrative access, with predefined roles (read-only, operator, policy-admin, super-admin) and the ability to create custom roles. We support SAML/OIDC-based admin authentication via HCT's institutional IdP.
- **Policy Hierarchy:** Our platform supports a hierarchical policy structure (global, regional, site-level, user-group-level) with policy inheritance and override capabilities, simplifying management and ensuring consistency.
- **Change Audit Trail:** We provide an immutable audit log of all administrative actions, including policy changes, user modifications, and configuration changes, complete with timestamps and administrator identity.
- **Real-Time Dashboards:** Strata Cloud Manager features pre-built and fully customizable real-time dashboards for security events, user activity, application usage, and the overall threat landscape.
- **Scheduled and On-Demand Reporting:** Our platform offers configurable scheduled reports (daily, weekly, monthly) and on-demand report generation for compliance, executive summaries, threat analysis, and incidents. Reports can be exported to PDF, CSV, and are accessible via API.
- **Alerting and Notification:** We provide configurable alerting thresholds with notifications delivered via email, webhooks, and syslog. The platform integrates with ITSM platforms for automated ticketing on critical alerts.
- **API Access:** A comprehensive RESTful API is provided for automation, integration, and custom reporting, covering policy management, user management, log retrieval, and configuration.
- **NOC / SOC View:** Our centralized dashboards are designed to provide NOC/SOC teams with complete awareness of threats, events, and network/internet activity.
- **Log Retention:** We offer high-capacity online log retention for up to 12 months, with options for extended archival. Log forwarding to third-party syslog, CEF, or SIEM solutions is fully supported.
- **Indicator of Compromise (IOC):** Our platform continuously scans security logs to identify suspicious traffic patterns and signs of compromise, presenting a prioritized list of potentially compromised hosts.
- **HTML5-based Web UI:** The Strata Cloud Manager console is a modern, fully HTML5-based web UI, with no Java, Flash, or ActiveX components.
- **3rd Party Integration:** We support integration with third-party systems via JSON and XML Web Services APIs.
- **(Optional ZTNA) ZTNA Dashboard:** For the optional ZTNA scope, the management console includes a dedicated ZTNA dashboard showing active sessions, application access summaries, connector health, user access patterns, and ZTNA-specific security events.

- **(Optional ZTNA) ZTNA Access Reports:** The platform generates detailed ZTNA-specific reports, including per-user access audit trails, per-application access summaries, and reports on denied access attempts and device posture failures.

3.8.11 ZERO TRUST NETWORK ACCESS FOR PRIVATE APPLICATION ACCESS (OPTIONAL)

Palo Alto Networks' optional ZTNA offering is a fully-featured, ZTNA 2.0 compliant solution that provides secure access to private applications for 200 designated HCT users.

Figure 9: ZTNA 2.0 architecture illustrating how the Prisma SASE cloud broker facilitates secure, cloud-brokered access to HCT's private applications via lightweight, outbound-only ZTNA Connectors.

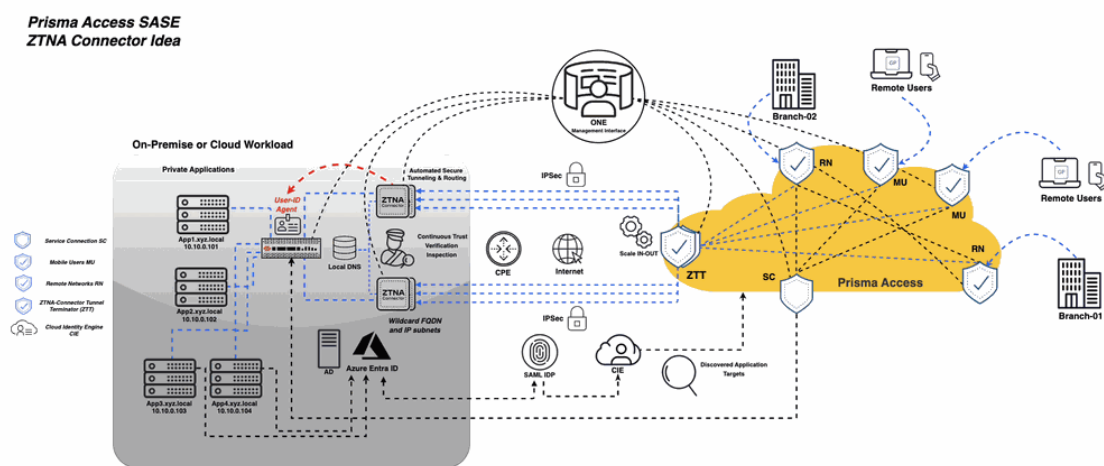


Figure 9: ZTNA 2.0 architecture illustrating how the Prisma SASE cloud broker facilitates secure, cloud-brokered access to HCT's private applications via lightweight, outbound-only ZTNA Connectors.

3.8.11.1 ZTNA Architecture Requirements

- **Cloud-Brokered, Zero Trust Architecture:** Our solution implements a true cloud-brokered architecture. Users never receive direct network-layer access to the internal network. Access is granted on a per-application, per-session basis, with continuous verification of identity, device posture, and contextual risk.
- **Lightweight ZTNA Connectors:** We use lightweight ZTNA Connectors (software appliances or virtual machines) deployed inside HCT's network (Primary DC, DR DC & Cloud). These connectors initiate outbound-only connections to the Prisma Access cloud, meaning no inbound firewall rules, DMZ exposure, or public IP addresses are required for the connectors, drastically reducing the attack surface.

3.8.11.2 Private Application Support

- **Supported Application Types:** Our ZTNA solution supports publishing a wide range of application types, including:
 - Web applications (HTTP/HTTPS) accessed via browser.

- Thick client / client-server applications.
 - RDP and SSH for remote access to servers and network devices.
 - VoIP/SIP-based applications.
 - Database client access (e.g., SQL Server Management Studio).
 - File share access (SMB/CIFS) to specific on-premises file servers.
- **Application Scalability:** The platform will support publishing a minimum of 100 private applications with the ability to scale to 300 applications without additional licensing costs during the contract period.
 - **Application Discovery:** Our platform supports automated discovery of private applications accessed by the designated 200 users during the migration phase, simplifying the process of identifying and publishing all necessary applications via ZTNA.
 - **Application Health Monitoring:** The platform continuously monitors the health and availability of all published private applications, alerting HCT and our support team if an application becomes unreachable via the ZTNA connector.

3.8.11.3 ZTNA Access Control and Policy

- **Per-Application Access Policies:** Access to each published application is controlled by a granular policy that evaluates user identity, group membership, MFA status, device posture (OS patch, EPP status, domain join, etc.), user risk score, geographic location, time of day, and device type.
- **Continuous Authorization:** Access is not a one-time gate. Our platform continuously re-evaluates the user's posture and risk score throughout the session. If a device's posture degrades (e.g., EPP is disabled), the ZTNA session is automatically downgraded or terminated in real-time.

3.8.11.4 ZTNA Inline Security

- **Threat Inspection:** All files downloaded from or uploaded to private applications through the ZTNA path are subject to full threat inspection, including malware scanning and sandboxing with Advanced Wild-Fire.
- **Full Visibility and Logging:** The platform provides complete visibility into all ZTNA session activity, including user identity, device, application accessed, session duration, and data volume. All ZTNA logs are forwarded to HCT's SIEM alongside internet traffic logs for unified analysis.
- **Anomaly Detection for Private Access:** Our UEBA engine extends to ZTNA sessions, detecting anomalous patterns such as access to applications outside normal working hours, unusually large data downloads, or access from previously unseen geographic locations.

3.8.11.5 ZTNA User Scope and Licensing

- **Initial User Scope:** The initial ZTNA deployment will cover 200 named users as specified, including IT administrators, senior staff, finance/HR staff, and designated contractors.
- **Scalable Licensing:** We will provide a scalable licensing model that allows HCT to increase the ZTNA user count in increments of 50 users at a pre-agreed, locked-in per-user rate for the contract duration.
- **Separated Licensing Costs:** In our commercial offer, we will clearly separate the licensing costs for the optional ZTNA scope from the base SASE internet security licensing, enabling HCT to award or decline the ZTNA scope independently.

3.9 APPENDIX-B

3.9.1 TECHNICAL REQUIREMENTS COMPLIANCE MATRIX

Palo Alto Networks confirms its ability to comply with all conditions and sub-clauses stipulated in the Scope of Work. The following matrix provides a summary of our compliance status for each domain. Detailed explanations are provided throughout Section 3 of this proposal.

Domain	Requirement	Compliance (Yes/No/Partial)	Remarks
3.	Technical Scope of Work	Yes	Palo Alto Networks Prisma SASE provides a comprehensive, unified solution that meets all technical scope requirements.
3.1	General Scope of Work	Yes	Our solution fully aligns with the general scope, offering a unified, cloud-delivered SASE platform with centralized management, AI-powered security, and consistent policy for all users and devices.
3.2	SWG Requirements	Yes	Prisma Access delivers a complete, best-in-class SWG with full SSL/TLS inspection, granular URL categorization, advanced threat prevention, and customizable user notifications.
3.3	FWaaS Requirements	Yes	Our FWaaS provides full NGFW-equivalent capabilities, including Layer 7 App-ID, inline IPS, and policy control based on IP, FQDN, device, and user identity.
3.4	RBI or Enterprise Requirements	Yes	Our Prisma Browser, a secure enterprise browser, provides superior security and user experience compared to traditional RBI, meeting all specified requirements for isolation and data exfiltration prevention.
3.5	AI-Powered Security Requirements	Yes	Our platform is infused with Precision AI™, delivering advanced capabilities for threat

Domain	Requirement	Compliance (Yes/No/Partial)	Remarks
			detection, UEBA, operational intelligence, and granular governance of Generative AI applications.
3.6	BYOD-Specific Requirements	Yes	We offer a multi-faceted approach for agentless BYOD security, including DNS redirection, PAC file support, certificate-based authentication, and our secure Prisma Browser.
3.7	Managed User Requirements	Yes	Our unified Prisma Access Agent provides all required SASE functions, including always-on connectivity, device posture assessment, and agent tamper protection for managed devices.
3.8	Platform Infrastructure and DEM	Yes	Our global, resilient, high-performance platform is backed by industry-leading SLAs for uptime, latency, and throughput. Our integrated ADEM provides deep experience monitoring.
3.9	Integration Requirements	Yes	Prisma SASE is designed with an open, API-first architecture for seamless integration with HCT's entire security ecosystem, including IdP, NAC, NGFW, EDR, SIEM, and UEM.
3.10	Management and Reporting	Yes	Strata Cloud Manager provides a single, intuitive console for all management, monitoring, and reporting functions, with customizable dashboards and comprehensive API access.
3.11	Managed Support Services	Yes	Palo Alto Networks, in conjunction with a certified local partner, will deliver all required managed support services, including 24x7x365 monitoring, incident management, and proactive maintenance.
3.12	Training and Knowledge Transfer	Yes	We will provide a comprehensive training and knowledge transfer program, including OEM official training and extensive shadowing, to ensure HCT's staff are fully enabled.
3.13	Documentation Requirements	Yes	We will deliver a complete set of detailed documentation, including architectural diagrams, design documents, runbooks, and a formal risk register, ensuring full operational readiness.
3.14	Service Level Agreements	Yes	We agree to the specified incident response and resolution SLAs and will provide a real-

Domain	Requirement	Compliance (Yes/No/Partial)	Remarks
			time SLA dashboard and formal monthly reports. Our mandatory escalation matrix ensures accountability.
3.15	Equipment Delivery or Project Timeline	Yes	We commit to the 3-month deployment and migration timeline, including a 30-day parallel operation period and a fully tested rollback plan.
3.16	Special Terms and Conditions	Yes	We agree to all special terms and conditions, including the 3-year contract duration, price lock, vendor certifications, data protection, and vendor lock-in mitigation requirements.
3.17	Project Management and Handover	Yes	Our Professional Services team will provide a dedicated PM and follow PMBOK standards, concluding with a formal Project-to-Operations Handover process.
3.18	Configuration Hygiene and Lifecycle	Yes	Our managed services include a rigorous quarterly configuration review and clean-up program to ensure the platform remains optimized and secure.
3.19	Compliance and Regulatory Requirements	Yes	Our solution is designed to meet UAE national regulations and international standards, including NIST, ISO 27001, and FERPA. We will provide all required compliance mapping and evidence.
3.20	Business Continuity and Disaster Recovery	Yes	We will provide a comprehensive BCP/DRP for the SASE platform, detailing our regional PoP resilience strategy and emergency bypass procedures.
3.21	Governance Framework	Yes	We will establish the proposed three-level governance framework and provide quarterly vendor performance scorecards against the defined KPIs.
3.22	Audit and Assurance	Yes	We agree to HCT's right to audit and will provide all necessary cooperation for audits, penetration tests, and joint security exercises.
3.23	Legal Protections and Contract Governance	Yes	We agree to the specified terms regarding limitation of liability, intellectual property ownership, termination rights, and transition assistance.
3.24	Proof of Concept (PoC) and Acceptance	Yes	We commit to conducting a comprehensive, 30-day PoC at our own cost to demonstrate all required capabilities and meet the specified success criteria.

Domain	Requirement	Compliance (Yes/No/Partial)	Remarks
3.25	Zero Trust Network Access for Private Application Access (Optional)	Yes	Our optional ZTNA 2.0 solution fully meets all architectural, security, and policy requirements for providing secure private application access.

3.10 TECHNICAL EVALUATION CRITERIA

Palo Alto Networks' proposal is designed to score exceptionally well against HCT's technical evaluation criteria. We offer the most complete, integrated, and AI-powered SASE platform on the market.

Criteria	Weightage	Palo Alto Networks' Compliance and Strengths
1. Company Profile		
1.1 Company Profile	10%	Excellent (10%): Palo Alto Networks is the undisputed global leader in cybersecurity and a pioneer of the SASE market. Our proposal provides a comprehensive overview of our extensive experience, global presence (including in the UAE), and ISO/IEC 27001:2022 certification.
1.2 Certifications & Partnerships	15%	Platinum / Premium (15%): As the principal OEM, we hold the highest-level certifications for all proposed products. Our direct access to R&D and support resources ensures unparalleled expertise and assistance.
1.3 Past Project Success & References	10%	6 and above (10%): We have numerous successful SASE deployments in the Higher Education and Federal sectors, both globally and within the Middle East. We will gladly provide more than six relevant references.
2. Team		
2.1 Team structure	10%	Strong offering (10%): We will provide a well-defined team structure, including a dedicated Project Manager and Resident Engineer, staffed by highly qualified and certified local and global experts. Detailed resumes will be provided.
3. Technical Criteria and Scope Compliance		
3.1 SASE Architecture & Design	15%	Excellent (15%): Our Prisma SASE architecture represents the industry's best practice for cloud-native security. It is unmatched in its integration, scalability, and AI-powered capabilities, providing a superior foundation for HCT's Zero Trust journey.
3.2 Managed Services	10%	Excellent (10%): Through our world-class Global Customer Services (GCS) and a certified local part-

Criteria	Weightage	Palo Alto Networks' Compliance and Strengths
		ner, we will deliver proactive monitoring, rapid incident resolution, and comprehensive maintenance and health checks as a fully managed service.
3.3 Scope Compliance	10%	100% (10%): Our proposal demonstrates full compliance with all technical features outlined in Sections 3.1 to 3.10, often exceeding the specified requirements.
3.4 Governance, Audit, Legal, BCP	10%	Full (10%): We fully comply with all requirements for configuration hygiene, compliance, BCP/DRP, governance, audit rights, and legal protections, offering a robust and transparent partnership framework.
3.5 SLA and Financial Commitments	5%	Full (5%): We accept the escalation matrix, measurement methodology, security efficacy SLAs, and liquidated damages clauses, demonstrating our commitment to service excellence and accountability.
3.6 PoC Results	5%	All criteria met (5%): We are confident that our Proof of Concept will meet or exceed all acceptance criteria defined in Section 3.24.2, including for the optional ZTNA scope, showcasing the tangible benefits of our solution.
Total Weightage	100%	

4. PROJECT MANAGEMENT

To maintain, run and deliver the project completely, MDS will appoint a Project Manager. MDS's Project Manager will be responsible for ensuring that the work has been completed in a timely manner, within our project scope, and meets the project requirements. Following are sample deliverables:

- Be the Single Point of Contact for all communications with the client.
- Oversee the entire project lifecycle.
- Setup a project review committee for periodic project audit.
- Know the project business objectives and build a clear plan to achieve them.
- Identify risks at the onset of the project and build a solid mitigation plan.
- Build, deliver, maintain & update project documents
 - Project Charter
 - Project Plan (including tasks, timelines, milestones, dependencies, resources)
 - Statement of Work
 - Communication plan
 - Resource Plan
 - Progress Report based on Progress meetings.
 - Minutes of all project meetings.
 - Escalation matrix.
 - Project Organization Structure
 - Stakeholders Register
 - Risk Register
 - Issue Log
 - Deliverables list
 - Handover/Signoff documents
- Coordinate and schedule the project activities.
- Ensure timely execution of MDS's responsibilities per the desired level of quality.
- Ensure optimum resource utilization.
- Manage & Coordinate between all MDS's teams and subcontractors (if any)
- Be the First escalation point for any delay in deliverables and quality issues.
- Escalate key issues to HCT at the end of each review.
- Hold Weekly / Bi-Weekly project progress/status meetings.
- Following up on action items.
- Manage Change requests.
- Following up and assuring deliverables are met.
- Ensure testing and training is completed with no open issues.

4.1 PROJECT METHODOLOGY

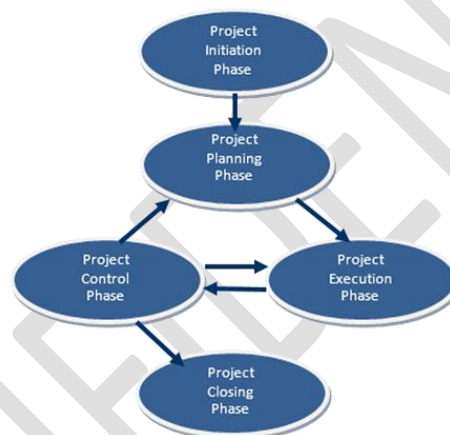
The following describes the Project Methodology employed by MDS. This methodology is based on recognized Project Management principles, drawing particularly on PMI's PMBOK.

Each project is divided into several processes or phases, each phase having its own identity and characteristics, the respective phases are:

1. Initiation Phase
2. Planning phase
3. Execution phase
4. Monitoring and Control phase
5. Project closing phase

A Project Schedule/Plan is compiled detailing activities within each phase with activity duration, start and end dates of each activity, and resources allocated to each of the activities.

The above diagram is applicable to the overall project, as well as to individual activities or key deliverables within a project



4.1.1 INITIATION PHASE

This is the starting phase of a project. Teams from both sides sit to discuss and finalize the objectives, scope, purpose and deliverables to be produced. Process, people and existing technologies can also be evaluated, and an appropriate change management program is developed to run alongside the implementation of the technology.

This provides insight of changes required, availability of personnel, level of change management that is necessary during the full lifetime of the project. These factors need to be considered in planning and executing a project on time and within budget.

The following items are discussed, agreed upon and documented in the Project Plan:

1. Project overview, goals and objectives
2. Project organization and structure, including roles and responsibilities of each project member

3. Project status meetings, frequency and participants
4. Project target milestones (key deliverables)
5. Project risks
6. Project logs
7. Quality Assurance
8. Change management procedures
9. Training and skills transfer
10. Maintenance and Support
11. Acceptance/sign-off of key deliverables

The duration of the implementation is dependent on a variety of factors including availability of resources and the level of complexity of the processes within the business.

4.1.2 PLANNING PHASE

The project planning phase results in the completion of the Project Schedule and resource planning and allocation. A Project Schedule form is compiled showing detailed activities, with durations and resource allocations. A critical path is normally shown if the project is resource driven, especially with time as a major constraint.

The following items are addressed in the planning phase:

- Project activity planning
- Project activity scheduling
- Organizational and resource planning
- Procurement planning
- Communications planning
- Risk identification, quantification and management
- Training and Skills transfer planning
- Quality planning

During the planning phase, decisions will be taken concerning which of various options are taken (for example project planning or labor and resources) based on the functionality required together with incremental cost, or possibly cost savings could be obtained. Decisions are also taken concerning sequencing of the business units to be implemented, of existing software that requires interfaces into the proposed solution, and of the efficiencies that are obtained by so doing.

4.1.3 EXECUTION PHASE

The project execution phase comprises the implementation of the items detailed under planning above. This is when design, engineering, testing and commissioning activities are executed. The following items are addressed in the execution phase:

- Project plan execution
- Project schedule execution
- Activity execution
- Quality assurance
- Project communications

4.1.4 CONTROL PHASE

The project control phase comprises of the measurement of activities against the project schedule, and the recording of any variances to these. Variances that could affect the project delivery negatively are given urgent attention (they are referred to the planning phase) and the necessary planning done, or actions taken to restore project delivery as originally planned, if possible.

The following items are addressed in the execution phase:

- Change control
- Schedule control
- Quality control
- Risk control
- Cost control
- Performance reporting

4.1.4.1 Project Change Control

The Project Manager, in conjunction with client, will be responsible for administering the Project Change Control Procedure.

Either party may request changes to the Statement of Work at any time. Since a change could affect the price, schedule, or other terms of the Agreement for this Statement of work, the Client Project Manager and the MDS Project Manager must approve each change before amending the Statement of Work and implementing the change.

This procedure will be used by Client and MDS to control changes to the Statement of Work and changes to any previously approved deliverables.

All Project Change Requests (PCRs) must be submitted in writing. They will describe the change and include whatever rationale and/or estimated effect the change will have on the Statement of Work. The Client Project Manager or MDS Project Manager, as appropriate, will review the proposed change. It is then accepted or rejected for submission to the other party. If rejected, the PCR is returned to originator along with the reason for the rejection.

The Client Project Manager and the MDS Project Manager will weigh the merits of the proposed change and approve it for investigation or reject it. MDS will investigate the request and the effect on the price, estimated schedule or other terms of the Agreement for this Statement of Work will be determined. The change will then be approved or disapproved for implementation.

Approved changes will be incorporated into the Statement of Work through written change authorizations (change orders) as appropriate.

MDS shall have no obligation to commence work in connection with any change until the fee and schedule impact of the change is agreed upon in a written Change Request Form signed by the designated Project Managers from both parties.

4.1.5 PROJECT CLOSING OR KEY DELIVERABLE ACCEPTANCE PHASE

The project-closing phase comprises activities to bring a project to completion. It is also applicable to the acceptance and sign-off of a key deliverable at the end of a phase within the execution of the project.

The following items are addressed in the acceptance/sign-off phase:

- Acceptance / agreement on completion of key deliverables
- Completion of outstanding documentation
- Project handover to client
- Timelines

Estimated Project Duration is subject to the following pre-requisites being fully completed:

- Work Order Signed & LPO received
- Hardware completely delivered on site
- Client provide all the necessary pre-requisite resources requested by MDS on time i.e. Rack Availability, Power connectivity, Network Connectivity, Internet, etc.
- Client Staff allocated for the project

4.1.5.1 Acceptance Criteria

MDS will notify Client in writing when a Material/Work which requires Client's acceptance under this Statement of Work is available for acceptance.

Within five working days after receiving MDS notice of availability, or before the end of the specified period, as applicable, Client agrees to notify MDS in writing of the acceptance of the Material/Work, or any deviations from the agreed upon:

- Acceptance criteria
- Anticipated test results, as applicable

If Client notify MDS of any deviations, MDS will correct them and re-notify Client that the Material is available for acceptance. Client agrees that MDS may repeat the above process until Client accepts the Material.

If Client does not notify MDS of any deviations within the time specified above or, if Client uses the Material productively, it will be deemed to have been accepted. Following the completion of all deliverables, the final status meeting and status report will confirm project closure. Any issues with the content of the final status report should be reported by Client within **one week of receipt**, otherwise the project will be considered closed.

4.1.5.2 Completion Criteria

Following the completion of all deliverables, the final status meeting and status report will confirm project closure. Any issues with the content of the final status report should be reported by Client within **one week of receipt**, otherwise the project will be considered closed.

MDS shall have fulfilled its obligations under this SOW when any of the following occur:

- MDS delivers the materials listed in each of the "Deliverables" sections.
- MDS terminates the Project in accordance with the provisions of the Agreement.

4.1.6 MDS STANDARDS AND COMPLIANCE

The project will adhere to the following standards and compliance where applicable:

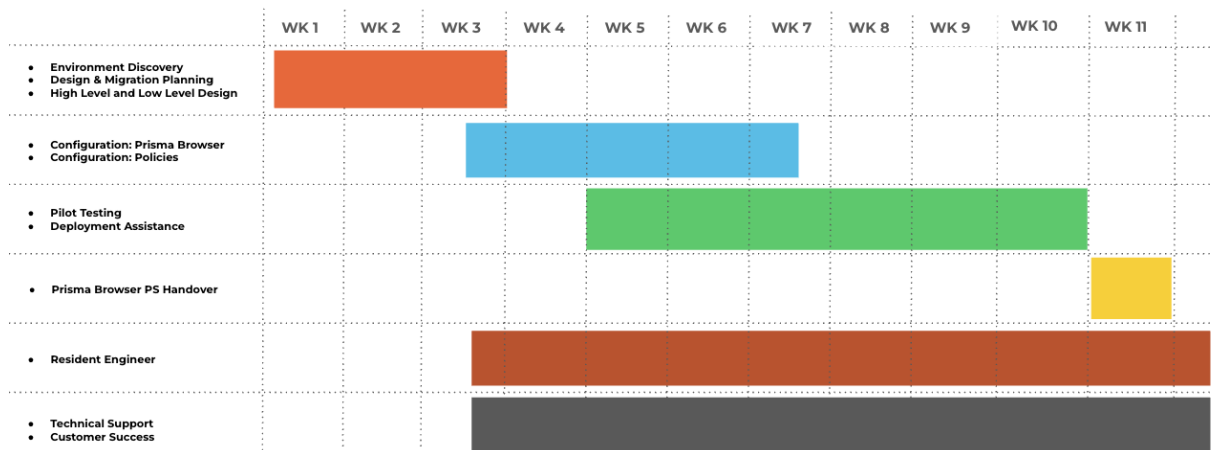
- PMI's PMBOK
- ITIL

4.2 RISKS

Risk	Impact	Owner	Mitigation
Rack Space & Power Requirements	H	HCT	Proper planning to assure availability prior to project roll out
Resource shortfalls	M	MDS/ HCT	Proper resource planning and validation.
Decision makers	H	HCT	Customer should assign a project manager for timely discussions and agreements on various project issues
Ill-defined project scope The general risk of an error or omission in scope definition	H	MDS/ HCT	Clearly define the scope on paper, at the onset of the project, agreed upon by both parties.
Inaccurate dependencies	M	MDS/ HCT	Dependencies dramatically impact the project schedule and costs. Clearly define the dependencies, at the onset of the project, agreed upon by both parties.

4.3 Migration and Project Delivery Plan

Indicative Timeline* - Migration and Deployment



* High-level plan dates are estimates until a planning workshop is held with all parties and the Project Plan is validated against PANW/Customer deliverables and dependencies.

To optimize costs and ensure a seamless operational transition, our migration and delivery approach employs a blended resourcing model as outlined in the indicative timeline. Palo Alto Networks Professional Services (PS) will lead the foundational phases, including Environment Discovery, Design & Migration Planning, and the initial baseline configuration of the Prisma platforms. Once the design phase is completed, a dedicated Resident Engineer (RE) will engage to participate in the core configuration alongside PS, and will subsequently take over to build on top of the initial configuration. The RE will handle continuous optimization, ongoing deployment assistance, and day-to-day operations through Pilot Testing and beyond. This strategy ensures that the environment is architected to best practices by PS experts, while the RE provides cost-effective, continuous optimization and localized technical support through the remainder of the deployment lifecycle.

4.4 TRAINING, DOCUMENTATION, AND KNOWLEDGE TRANSFER DETAILS

4.4.1 FORMAL TRAINING OFFERINGS

Palo Alto Networks Education provides comprehensive, instructor-led training delivered by Authorized Training Partners to ensure your operational teams are fully equipped to protect your applications, remote networks, and mobile users.

- **Prisma Access SSE: Configuration and Deployment**
 - **Duration:** 4 days
 - **Format:** Lecture and hands-on labs
 - **Overview:** Introduces operational deployment of Prisma Access Secure Access Service Edge (SASE). Enables teams to deploy, configure, maintain, and troubleshoot Prisma Access via Strata Cloud Manager.
 - **Key Modules:** Prisma SASE, Architecture, Strata Cloud Manager, Service Connections, Remote Networks, Mobile Users, ZTNA Connector, Prisma Access Browser, and Autonomous Digital Experience Management (ADEM).
- **Prisma Browser**
 - **Duration:** 1 day
 - **Format:** Lecture and hands-on labs
 - **Overview:** Focuses on securing internet and private applications using a secure enterprise browser. Enables teams to deploy, configure, maintain, and troubleshoot Prisma Browser using Strata Cloud Manager.
 - **Key Modules:** Architecture, Policy, Monitoring, Logging, Troubleshooting, Prisma Browser Extension, and Prisma Browser Mobile.

(Target Audience for both courses: Security Engineers, Security Administrators, Security Operations Specialists, Security Analysts, and Network Engineers.)

4.4.2 DOCUMENTATION DELIVERABLES

Palo Alto Networks Professional Services will develop and deliver extensive documentation throughout the project phases based on Palo Alto Networks standard templates.

- **Planning & Discovery Phase:**
 - **Project Kickoff Document:** Defines key stakeholders, project scope, communications plan, and risk/issue methodology.
 - **Initial Project Plan:** Provides a baseline plan detailing tasks and dependencies.
- **Design Phase:**
 - **Technical Requirements Document (TRD):** Captures deployment scenarios, network integration points, design requirements/objectives, SaaS application requirements, data security policies, and use cases.

- **High-Level Design (HLD) Document:** Details the architecture, system hardening, foundational capabilities, and overarching deployment/validation strategies.
- **Low-Level Design (LLD) Document:** Provides the granular configuration requirements for security, networking, routing, and component integration (e.g., Prisma Access, Prisma Browser, NG-CASB).
- **Post-Deployment:**
 - **As-Built Configuration Document:** Delivered at the conclusion of the deployment, this document serves as a snapshot of the actual implementation and all configuration changes made within the service parameters.

(Note: Custom formal training manuals, workbooks, lab guides, or updates to the customer's existing documentation are explicitly excluded from the standard Professional Services engagement.)

4.4.3 KNOWLEDGE TRANSFER

To ensure a smooth transition to your IT operations and security teams, Palo Alto Networks provides dedicated Knowledge Transfer (KT) sessions after the completion of the planning, configuration, and validation phases. These sessions include a detailed walkthrough of the as-built environment and practical information on how to manage and operate the deployed solution.

- **Prisma Access Advanced Deployment:** Two (2) Knowledge Transfer sessions.
- **NG-CASB Advanced Deployment:** Two (2) Knowledge Transfer sessions.
- **Prisma Browser and Privileged Remote Access (PRA) Advanced Deployment:** One (1) Knowledge Transfer session.

Session Parameters:

- **Duration:** Up to four (4) hours per session.
- **Capacity:** Maximum of eight (8) participants per session.
- **Format:** Remote delivery.

5. PROJECT GOVERNANCE METHODOLOGY

To control the project in terms of communicating, implementing, monitoring, and assuring the policies, procedures, organization structures, and practices associated with a given program, it is imperative to build a Project Governance Plan.

The purpose of the Governance Plan is to describe the specific roles and responsibilities of program management and its stakeholders, focusing primarily on authority level and decision-making structures. The governance of a project is coordinately performed through several bodies, mainly: the project team, and the Steering Committee.

The proceeding parts of this section will describe MDS's project governance methodology.

5.1 PROJECT TEAM

- MDS and HCT will each name a responsible Program Sponsor at the beginning of the project. The Program Sponsor is the overall champion and owner of the project and his/her role is to glue all the stake holders together, approve the project charter, prioritize the project within their organization.
- MDS and HCT will each name a responsible Program manager at the beginning of the project. The Program manager is responsible for the budget, the schedule, and the entire performance of the project.
- Technical project leaders from each side are to be named for the individual technical areas at the beginning of the project. They are responsible for the technical progress of the project and lead the respective project teams. In regular meetings they will inform the project steering committee about the project progress, and should the situation arise, report problems requiring decisive action.
- The project team and the steering committee composition will be finalized as part of the project kick-off activities.
- Weekly Project Progress Review Meetings will be conducted at the project management level. Attendants to these meetings will be: HCT project manager, MDS project manager, HCT Lead Solution Architect, MDS Lead Solution Architect
- Other persons may attend as needed: MDS different Solution Architects, HCT subject matter experts, HCT process owners
- The agenda of the weekly project progress review will include: schedule progress, planning for next tasks, resources management (including entrance permits processing and security clearance), dependencies on HCT resources, tasks and facilities; risk management, issues management, change requests.
- Additional meetings at the project team level will also happen before and after important milestones. The preceding meetings will help to assure readiness of resources, needed facilities, and coordination. The subsequent meetings will help to check results. The selection and planning of these meetings will be jointly done by HCT and MDS during the project delivery at an adequate date.

An initial list of these important milestones includes design of each technology, implementation of each technology, UAT (user acceptance testing) of each phase, and go-live.

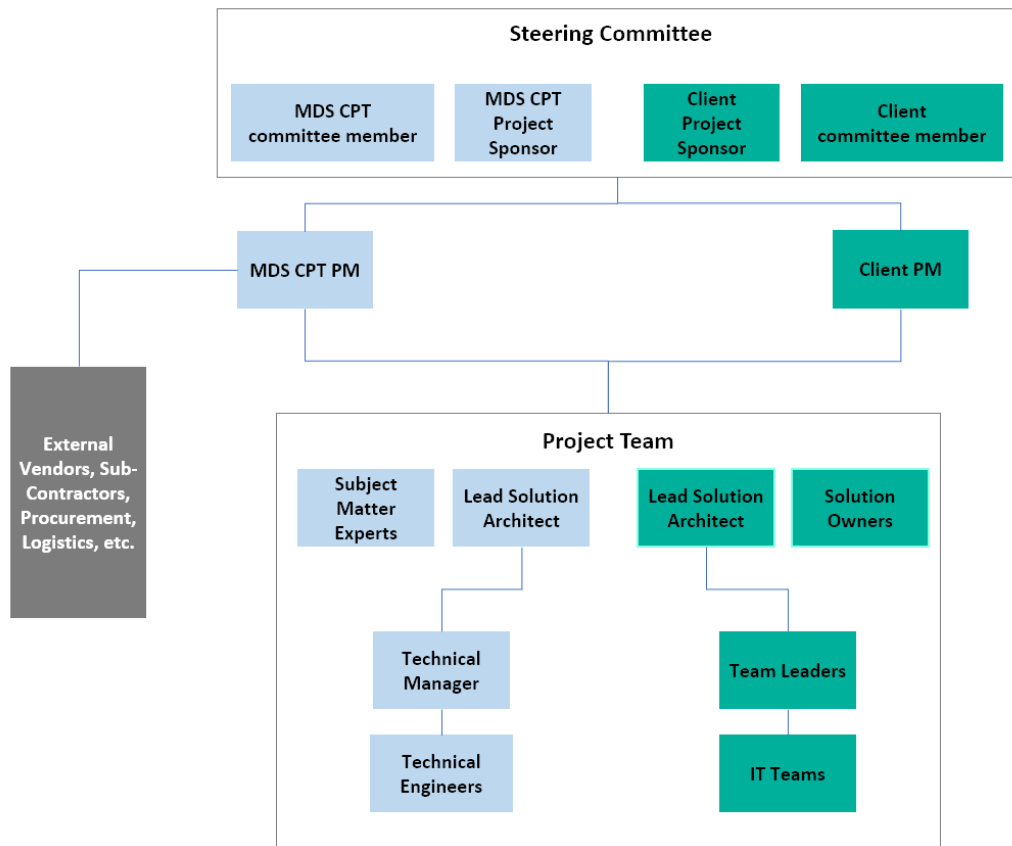
5.2 STEERING COMMITTEE

- The steering committee will/may consist of: HCT Program manager, MDS Program manager, HCT project sponsor, HCT decision making managers, MDS project sponsor, and MDS decision making manager.
- HCT should assign as project sponsor an empowered, high rank official with decision making power to ensure collaboration of the required HCT team.
- The responsibilities of the Steering Committee are:
 - Regular monitoring of project progress against high-level plan
 - Ensuring project delivers against strategic objectives
 - Managing escalated issues
 - Managing escalated risks
 - Budget decisions
 - High level staff communications
 - Informing project team of strategy and policy changes that will affect the project delivery
 - Take decisions on matters that cannot be solved in project teams
 - Policy decisions
 - The Steering Committee can also decide a variation of the Statement of Work if so jointly agreed
- The steering committee will meet monthly or when necessary to decide on matters that were escalated from project leadership.

In the absence of a member of the Steering Committee, the relevant party shall appoint as soon as possible, an alternate member of the Steering Committee, which will have the same authority as the absent member of the Committee.

5.3 PROJECT TEAM ORGANIZATION

Following is the general project team organization. Specific organization is built at the beginning of the project after assessment of the project needs, scale and complexity.



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5.4 MEANS OF COMMUNICATION

Utilization of a formal process will ensure timely and accurate communication between MDS and HCT organizations. There are two key vehicles for providing this communication:

1. A weekly status report and
2. A weekly status meeting.

All team members will report progress and status weekly. This information, and the analysis of it, will be summarized into a project status report for HCT and MDS management. This report contains the following information:

1. Significant problems/actions taken
2. Progress summary
3. Accomplishments
4. Planned accomplishments
5. Schedule status
6. Issues log

The weekly status meeting of the project management team will be used to review the project status and open issues in the status report. The HCT and MDS project management team must commit to having this meeting and making it effective. The keys to making the meeting effective are:

- Have the meetings often enough so that timely information is available
- Focus the meeting on major issues or areas where management action or intervention is required
- Include project staff involved in principal focus of each week's activities
- Make sure that the information provided is accurate and objective

5.5 MANAGING SUBCONTRACTORS

For projects where MDS has to manage several Subcontractors (which could include manufacturer's services teams) under it implementing the different parts of the solution MDS will put in place a system of processes and protocols to make sure the different teams work together to deliver the overall solution, including:

- Define a communication matrix and make sure information (status and progress reports, gathered technical info, timelines/deadlines, dependencies, etc.) is shared between all teams in a timely manner and at the appropriate level.
- MDS Lead solution architect will work closely with the Subcontractors architects to make sure the designs for the different solution are aligned together.
- MDS resources will be shadowing the Subcontractors resources to make sure work is done as per the agreed scope and schedule.
- Designate SPOCs for each Subcontractor/vendor to communicate with MDS PM.

- Escalation matrix and procedures will be defined for each Subcontractor.
- Regular Internal meetings between MDS and the different Subcontractors to make sure all parties are aligned.
- MDS will define templates for all documentation issued by the Subcontractors.
- MDS Quality team will review all documentation submitted from the Subcontractors to make sure they comply with the agreed standards.

5.6 ESCALATION PROCESS

Timely resolution of issues is critical to maintaining project control. The purpose of the escalation process is to help ensure that issues are identified and resolved quickly.

The escalation process provides a mechanism to alert Project Managers and other management personnel about issues not being resolved. The MDS Project Manager will maintain an Issue log to monitor and control issues and escalations, for the sake of project efficiency.

Either MDS or HCT may escalate a project issue as follows:

1. Raise the issue initially to the MDS and HCT Project Managers.
2. If not resolved at this level within 24 Hours, an issue report will be generated and the issue will be escalated to the MDS and HCT Project Sponsors.
3. If the issue cannot be resolved within a predetermined period (3 business days) or falls outside the authority of the MDS and HCT Project Sponsors, it will be escalated to the Project Steering Committee.

Certain issues may need to be escalated to the Principal Vendors for resolution.

5.7 RISK/ISSUE MANAGEMENT

Risk analysis and risk management are important factors in achieving a successful implementation effort.

Each risk nominally goes through the below functions sequentially, but the activity occurs continuously, concurrently (e.g., risks are tracked in parallel while new risks are identified and analysed), and iteratively (e.g., the mitigation plan for one risk may yield another risk) throughout the project life cycle.

Function	Description
Identify	Search for and locate risks before they become problems.
Analyse	Transform risk data into decision-making information. Evaluate impact, probability, and timeframe, classify risks, and prioritize risks.
Plan	Translate risk information into decisions and mitigating actions (both present and future) and implement those actions.
Track	Monitor risk indicators and mitigation actions.
Control	Correct for deviations from the risk mitigation plans.
Communicate	Provide information and feedback internal and external to the project on the risk activities, current risks, and emerging risks. Note: Communication happens throughout all the functions of risk management.

5.8 PROJECT CHANGE MANAGEMENT

During the project either party may request in writing additions, deletions, or modifications to the services described in this Statement of work ("change").

Since a change could affect the price, schedule, or other terms of the Agreement for this Statement of work, the Client Project Manager and MDS Project Manager must approve each change before amending the Statement of Work and implementing the change.

MDS Project Manager, in conjunction with the Client Project Manager, are responsible for administering the Project Change Management Procedure.

MDS shall have no obligation to commence work in connection with any change until the fee and schedule impact of the change is agreed upon in a written Change Request Form signed by the designated Project Managers from both parties.

Upon a request for a change, we shall submit the change on our standard change Request Form describing the change, including the impact of the change on the project schedule, fees and expenses.

The Change Management Process that will be employed is defined below:

- Identify and document
- Assess impact and prioritize
- Estimate required effort
- Approve/Disapprove
- Assign responsibility
- Monitor and report progress
- Communicate change resolution

5.9 QUALITY MANAGEMENT APPROACH

The purpose for managing quality is to validate that the project deliverables are completed with an acceptable level of quality. Quality management assures the quality of the project deliverables and the quality of the processes used to manage and create the deliverables.

The quality management approach is about to ensure that all personnel in the project "do the right things, right, the first time" thus avoiding misunderstandings of what to produce and how, resulting in productivity losses, quality losses and schedule delays.

MDS works with the client early in the project to prepare a quality plan that ensures the above goals are achieved, by:

- Identifying the client's Quality Objectives.
- Helping client express quality expectations in objective, quantitative terms.
- Balancing needs and expectations of clients and stakeholders with cost, schedule, and professional standards.
- Developing an effective plan and processes, including quality assurance and quality control procedures, to achieve objectives. Consider risk/hazard factors and complexity of the project and adapt processes to provide the requisite level of quality.

Quality management consists of several different elements. For MDS, the key elements of a quality management plan are:

1. The quality standard that needs to be met. This is the most important element and establishes what "success" looks like for quality.
2. How quality will be assured (quality assurance). This is the approach that will be taken during the execution of the project to ensure that achievement of the quality standards is "built in" to the project.
3. How quality will be ensured (quality control). This is the work that will happen toward the end of the project to measure actual performance of the deliverables against the standard to identify any variances and take corrective action if necessary.

5.9.1 QUALITY ASSURANCE APPROACH

Quality assurance is concerned with ensuring that the project execution meet the quality standards.

Later in the project, quality control will be used to find any failures to meet the standard, but it is much cheaper and easier to try and ensure that the work is undertaken in such a way as to prevent problems from occurring in the first place. This may consist of any or all the following:

- Processes and best practices that establish how to go about the work in such a way that common problems are avoided.
- Reviews that occur at regular intervals to try and capture any potential problems before they become "built in" to the solution.
- Policies that must be complied with to try and prevent problems from occurring.

- Reporting, approval and/or audit requirements to validate the work at various stages.

The following tool can be used for tracking and assuring quality:

- The project processes subject to quality assurance.
- The quality standards and stakeholder expectations for that process.
- The quality assurance activity – such as a quality audit or reviews - that will be executed to monitor that project processes are properly followed.
- How often or when the quality assurance activity will be performed.

5.9.2 QUALITY CONTROL APPROACH

Quality control is concerned with the measurement of actual results to validate that the quality standard has been met. It occurs at the end of the project when there are actual deliverables to manage against but should be planned for during the rest of project planning.

Project execution is a dynamic process. The Project delivery Team must communicate, meet on a regular basis, and adapt to changing conditions. The Quality Management Plan and Project Plan may require modification to ensure that project objectives are met.

The following tool can be used for quality control:

- The major deliverables of the project that will be tested for satisfactory quality level.
- The quality standards and the correctness and completeness criteria established for the project deliverable.
- The quality control activities that will be executed to monitor the quality of the deliverables.
- How often or when the quality control activity will be performed.

5.9.3 QUALITY ROLES AND RESPONSIBILITIES

Role	Responsibilities
Project Sponsor	Approve each project stage per framework checkpoints.
Project manager	Documenting client expectations and consensus quality management objectives at a project-level. Utilizing the expertise of the project delivery teams to determine the procedures necessary to achieve the target level of quality. Working with the client early in the Project Scope and Client Requirements Definition Process to determine client needs, and refining those requirements considering schedule, scope, and other constraints. Assure practice of quality control measures and communications.
Technical Lead	Performing an active role to ensure the client's quality objectives are clearly articulated. Monitoring the deliverables of the teams.
Delivery Team	Delivering a quality project. Monitoring the quality of their own work. Keeping the commitments for completion of their portion of the project.

5.9.4 PROJECT MONITORING

The project monitoring process within MDS is handled through weekly/ Bi-weekly projects progress reports (depending on the project duration) to MDS senior management as well as to HCT. These reports are generated from the projects schedules updated by MDS project managers and include the project overall status, performed activities in the past period, planned activities for the period ahead, achievements & milestones, project risks along with their mitigation plans & any outstanding issues. The project Governance & Quality Assurance processes within MDS is handled by MDS PMO, through Internal Project Management Audits to make sure that the projects are complying with MDS predefined standards and methodologies and also to make sure that HCT requirements are met.

5.9.5 PERFORMANCE TESTING

The performance testing & QA approach within MDS is handled in the Design & Planning Phase and the Stabilization & testing Phase as follows:
Creation of the acceptance criteria document during the Design & Planning Phase to include all the recommended detailed testing activities for each of the project deliverables. This document should be submitted to HCT for review & approval before the start of the implementation activities to set the expectation from the proposed solution.

During the Stabilization & Testing Phase, MDS technical team leader is responsible for performing the QA activities & system testing on the implemented solution by checking its health status, functionality and performance and overall to make sure that it is meeting HCT requirements as per the approved SOW and analysis document.

After the successful QA & system testing, HCT proceeds with the User Acceptance Testing using the approved acceptance criteria document to accept the project deliverables.

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6. MDS PROFILE

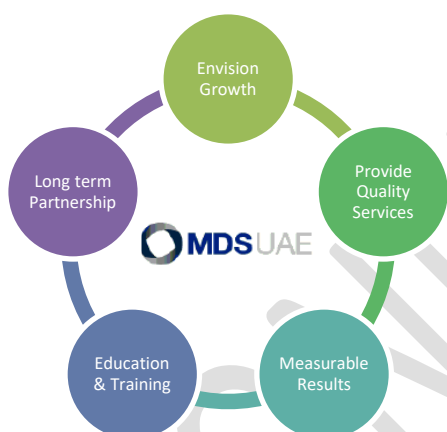
VISION "Where to"

To be the most reliable and respected partner in building a sustainable economy driven by advanced digital technologies.

MISSION "How"

To provide organizations, in public and private sector, the most compelling IT system integration and customer experience possible.

6.1 INTRODUCTION



CPT, part of Midis Group, is the leading Information Technology & Cybersecurity Company in the UAE and the Middle East. MDS has been serving the region for over 27 years and has employed over 2000 professionals. MDS has been nominated by IDC as the number one IT Service Company in the UAE for three consecutive years.

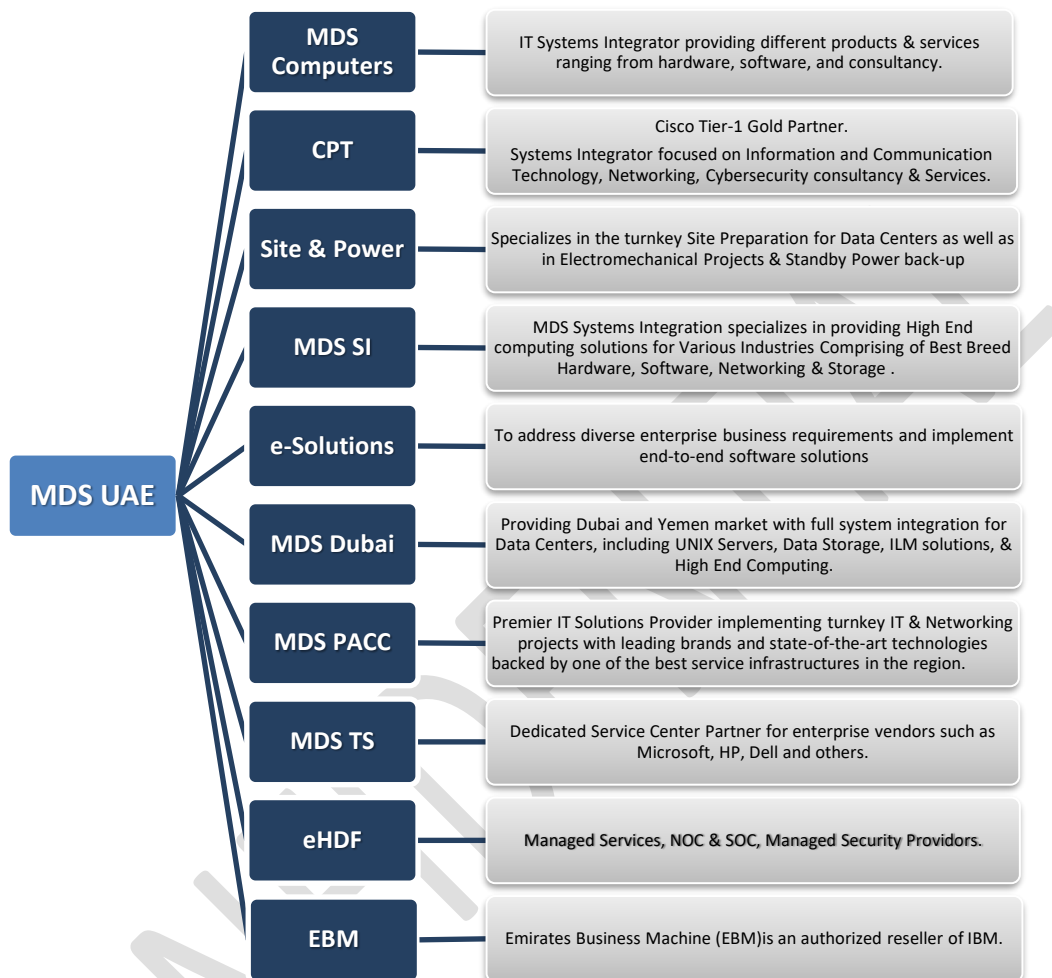
Our organizational structure and financial capability make us an ideal partner for the region's leading corporations who share our vision of explosive growth as well as for clients who are actively seeking partners with proven expertise and high commitment to establish their business in this market.

6.2 THE MDS UAE GROUP OF COMPANIES

As the leading system integrator, MDS has executed a huge number of turnkey IT projects in partnerships with global IT giants under the umbrella of MDS UAE group of companies. MDS has acquired vast and varied expertise needed to handle such projects with efficient and professional project management, customer management, and vendor management. MDS takes full responsibility of each project it undertakes and guarantees customer satisfaction.

Fast, responsive and attuned to customer needs, MDS is at the leading edge of IT innovation. We can help clients evolve existing IT Solutions and Services to better align with business processes to achieve

greater efficiency and productivity. Our primary goal is to deliver quality products and services that consistently exceed customer expectations while meeting their budgets and schedules.



The company's organizational structure and financial capability make it an ideal partner for both, leading companies who share MDS's goal to expand the market it serves, and for clients who are actively seeking suppliers that offer expertise and highly focused commitment to their markets.

6.3 MIDIS GROUP

With more than 3500 professionals, over 50 world leading IT brands & Suppliers, and 30 years of uninterrupted growth, the Midis Group is an International organization consisting of over 150 companies across the emerging markets of the Middle East, Central East Europe and Africa.

The group is known for its advanced offering of Advanced services, distribution, system integration, Software and hardware products along with Data Center consultancy, infrastructure and standby power solutions.

Across borders and languages, the Midis Group affiliates operate as partners in focused profit centers. Backed by the Group's financial strength, they strive for the best in market effectiveness and client satisfaction.

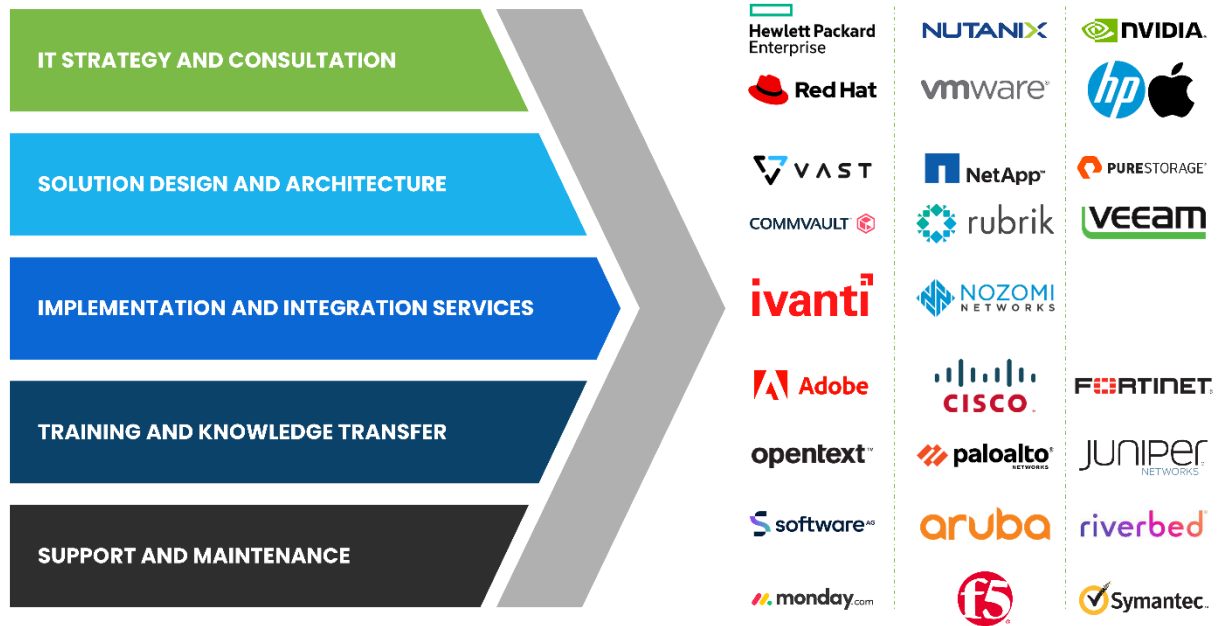
6.4 PRINCIPLES

MDS is always seeking the highest level of service and knowledge by constantly engaging with our partners. At MDS we are continuously training our Sales and Technical teams as we believe that "Experience" can't be acquired or conquered but only earned through years of hard work. Progress and excellence come from the intelligent use of such experience. As a high integrity partner, we build our business around:

- **Entrepreneurship**, Maintains flexibility, competitiveness, and initiative
- **Honesty**, Trust and mutual benefit make the strongest business partnerships
- **Expertise**, the knowledge sector in our regions requires skill and experience
- **Long term success**, the strongest vendors and clients endure

7. PARTNERSHIPS

Key Partners / Ecosystem



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7.1 PARTNERSHIP CERTIFICATES

7.1.1 PALOALTO PARTNERSHIP CERTIFICATE



February 27, 2025

To Whom It May Concern:

On Behalf of the Global NextWave Partner Program team at Palo Alto Networks, this is to inform you the partnership between Palo Alto Networks, Inc. and:

Mideast Data Systems L.L.C

Abu Dhabi, Non-U.S.

United Arab Emirates

At the time of this communication, **Mideast Data Systems L.L.C** is classified as a Solution Provider at the **Platinum Innovator** level partner respectively in the Palo Alto Networks NextWave Partner Program and is in good standing.

Mideast Data Systems L.L.C is also currently set with the following Specializations : **Hardware Firewall;Software Firewall**

Thank you,

A handwritten signature in black ink, appearing to read "Michael Khoury".

Michael Khoury
VP, Global Channel Programs
Palo Alto Networks

8. ENGINEERS' PROFILES

SIRAJUDHEEN NK Abu Dhabi-UAE ▪ Date of birth: 31 MAY 1987	Contact Tel : +971 526237968 E-mail : Sirajudheen@cptauh.ae
Professional Certifications	
     • CCIE Security (Cisco Certified Network Expert – CCIE#60452) • CCNP R&S (Cisco Certified Network Professional) • CCNA R&S (Cisco Certified Network Associate) • F5-CA (f5 certified system Administrator) • HCDP (Huawei Certified Data Professional)	
Work Experience	
• Senior Network and Security Engineer. Feb 2021- Till Date – MDS CPT – IT Solutions. UAE • Senior Network and Security Engineer. June 2013- Feb 2021 - AI MARIA MIDDLEEAST TECHNOLOGIES. UAE • Project Engineer. April 2012 – April 2013 – AGC Networks Ltd. INDIA • Network Engineer. Dec 2010 – March 2012 – Linknet Solution Pvt Ltd. INDIA	
Technical Skills	
• Implement and configure the network and security component as per the design and best practices • Network Security, Routing & switching, Reverse Proxy • Application Load Balancer, Web Application Firewall, Access Policy Manager • VLAN, Inter VLAN Routing, STP/RSTP/MSTP, VTP, Link Aggregation, Stacking, VSS • Redundancy Protocols VRRP, GLBP, HSRP • Routing Protocol RIP,EIGRP, OSPF, BGP • IPSEC, SSL, GRE, GRE over IPSEC, Dynamic VPN, Cisco DMVPN • Email Security – TrendMicro , Cisco ESA • Endpoint Security – TrendMicro, Fort client EMS • Firewall Policies/ACL, NAT, UTM (Antivirus, Anti-spam, Deep Inspection, Web filtering) and IPS • Network access control (NAC) – Cisco ISE	

Hands On Experience

- Cisco ASA with Firepower and FMC
- Fortinet Firewalls, FortiAnalyzer, FortiManager, Forticlient EMS.
- Cisco ISE and Cisco ACS - (SNS-3415,3495)
- BiG-IP LTM, APM, ASM
- Juniper EX series switches, SRX firewall, SA SSLVPN Box
- Cisco Nexus Switches
- TrendMicro Email Security
- Cisco Routers (39xx, 29xx, 19xx)
- Cisco Campus Switches (65xx,45xx,36xx,38xx, 29xx)
- IBM Qradar – SIEM
- AVI (NSX-ALB) Load Balancer, WAF.
- Trend Micro Deep Discovery Inspector,
- Trend Micro Endpoint Security Office scan
- Huawei and HP switches

Major Projects

- Confidential Customer: DC Upgradation Project: Successfully upgraded and redesigned the network of a customer Datacentre with F5 LTM-APM-ASM, Juniper SRX3400 Firewall, Cisco ASA5545 with IPS, Cisco Routers, and Switches
- Confidential Customer : Successfully completed F5 LTM, ASM and APM Implementation
- Royaljetgroup: Successfully completed the Fortinet Firewall Migration, Cisco ISE NAC Implementation with Wired&Wireless Dot1x Authentication, MAB and Guest Portal, TrendMicro Email security, Endpoint security, Deep Discovery Inspector, AVI NSX-ALB, Qradar SIEM, FortiManager, FortiAnalyser etc
- Confidential Customer: F5 BiG-IP LTM and ASM Implementation.
- British School : Implementation of Fortinet 601E firewall
- GCAA : Hardening of existing LTM Configuration and Configured APM for securing the application via TFA, Google Captcha, Host check and Published the internal Application via WebTop
- Royal Group : Successfully Implemented F5 LTM for Oracle Application
- Etisalat: Successfully Deployed F5 VE for Etisalat private Cloud.
- Shamal group : Successfully configured Cisco ASA with Firepower service
- Led the Datacentre Physical Migration of MOCCA and Royaljetgroup.
- Confidential Customer : Successfully completed the Project of SRX Implementation, Juniper SA, Cisco ACS
- SIPCO: ASA firewall Implementation with Firepower Service and FMC.

Formal Education

- B-tech (Bachelor of Technology) Electronics & Communication, NSS College of Engineering, Palakkad, Calicut University, India.

Personal Details

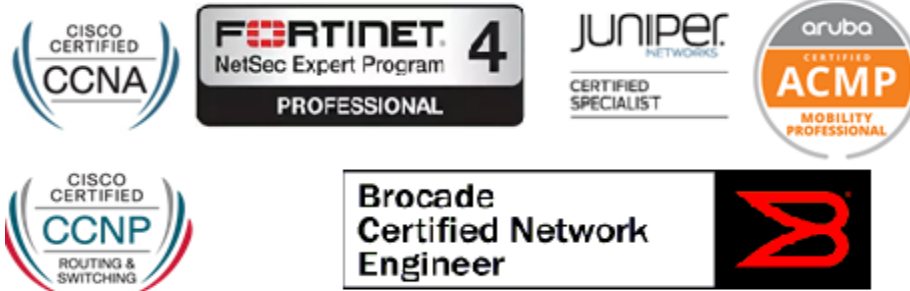
- Date of Birth: 31/05/1987
- Gender, Marital: Male, Married
- Country of residence: UAE
- Nationality: Indian
- Driver's License: Holding valid UAE and Indian Driving License
- Languages known: English, Malayalam
- Passport No: T-2797693

CONFIDENTIAL

RAMI YOUNIS

Abu Dhabi-UAE

Professional Certifications



- Cisco Certified Network Professional (CCNP)
- Cisco Certified Network Associate (CCNA)
- Juniper Network Certified Professional – R&S (JNCIP-ENT)
- Juniper Network Certified Specialist – R&S (JNCIS-ENT)
- Juniper Network Certified Specialist – Security (JNCIS-SEC)
- Juniper Network Certified Associate Junos (JNCIA-Junos)
- PaloAlto Networks Accredited Configuration Engineer (ACE)
- Brocade Certified Network Engineer (BCNE)
- Aruba Certified Mobility Professional (ACMP)
- Certificate of Attendance Advanced Junos Security (AJSEC)
- Certificate of Attendance Junos Security (JSEC)
- Fortinet NSE-4 Network Security Professional
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Technical Skills

- IP/Security Network Design, Configuration, and Operation
- VLAN, Inter VLAN Routing, STP/RSTP/MSTP, VTP, Link Aggregation, Stacking
- IPSEC, SSL, GRE, GRE over IPSEC, Cisco DMVPN
- Wireless networks design, implementation and troubleshooting
- Firewalls Configurations, HA, Security Policies, VPN, UTM
- Vulnerability management & CCM solution
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Hands on Experience

- Cisco ACI
- Cisco SD-Access
- Meraki
- Cisco ISR Routers
- Cisco Catalyst Switches
- Cisco ASA Firewalls
- Cisco Wireless LAN controllers
- Palo Alto Firewalls
- Aruba Wireless LAN controllers
- Aruba Switches

- Fortinet FortiGate firewalls
- Juniper EX series switches,
- Juniper SRX series firewalls
- Juniper Junos-Space Management

Major Projects

-
- UAE University: Full network implementation for LAN based on Cisco VSS (Cisco 9500 and Cisco 9300 Switches) and Wireless LAN based on Cisco WLC 5520
- Ministry of Presidential Affairs (MoPA): Juniper Fusion implementation based on Juniper EX9200 and Juniper EX4300
- BlueWaters Island: WLAN design, full configurations and implementation for Core switches, Server Farm Switches, Wireless LAN controller, Data Center Firewall.
- Burj Al-Arab: Switches Replacement, Full Configurations and installations for new Core switches (Brocade MLX-8), Distribution switches (Brocade ICX6610) and Access Switches (Brocade ICX6450).
- Madinat Jumeirah: Switches Replacement, Full Configurations and installations for new Core switches (Brocade MLX-8), Distribution switches (Brocade ICX7750) and Access Switches (Brocade ICX6450), Brocade Network Advisor for network management.
- Abu Dhabi University: Complete Network setup for Abu Dhabi University (Abu Dhabi Campus) for Core Switches (Juniper EX9208), Access Switches (Juniper EX4300) and DC firewalls (Juniper SRX4300).
- YUM Restaurants International: Cisco Routers and Switches Installation and configuration
- Baron Hotel in Iraq: Complete network Configuration for Core Switches (Brocade ICX7750), Distribution Switches (Brocade ICX6610) and Access Switches (Brocade ICX6450).
- Abu Dhabi Media Company (ADMC): New site Switches Installation (Brocade ICX6450).
- Jumeirah Beach Hotel (JBH): IDFs switches replacement (Brocade ICX6450 and Juniper EX4300).
- Jumeirah Group: Firewall Installation and Configuration for CCTV network (Juniper SRX240).
- Abu Dhabi National Oil Company: Physical security network implementation based on Juniper Networks switches and firewalls.
- Al-Dar Academies: Full Project implementation for two schools in Abu Dhabi for Network (Juniper EX4600, EX4300 and EX3300), Security (Juniper SRX1500) and WLAN (Aruba Networks),

Abdallah M D Yousef

Abu Dhabi-UAE • Date of birth: 7 July 1994

Contact

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Professional Certifications



- **NSE7** (Network Security Architect “Enterprise Firewall” – **DkX63Mu40E**)
- **PCNSE** (Palo Alto Networks Certified Network Security Engineer – **PY6B0PX0WW60FR96**)
- **NSE 4** (Network Security Professional – **wxAM8nQHrK**)
- **CCNA R&S** (Cisco Certified Network Associate – **CSCO13336444**)
- **PSE Strata** (Palo Alto Networks Security Engineer – Strata Professional - **YCNMNBTCB1Q41N3**)
- **Sophos Certified Engineer** (ET00 – Fundamentals, ET01 - Sophos Central Overview, ET80 - XG Firewall v17.0)

Work Experience

- **Senior Network and Security Engineer.** August 2021- Till Date – MDS CPT – IT Solutions. UAE.
- **Information Security Engineer.** Oct 2018- July 2021 - Future Technology Systems Co. Kuwait.
- **Network Engineer.** Feb 2018 – Sep 2018 – Future Tech. Kuwait.

Technical Skills

- Design, Implement and configure the network security component as per the design and best practices.
- Next Gen. Firewall Policies/ACL, Src& Dist NAT, UTM (Antivirus, App Control, Web filtering) and IPS.
- Network Security, Routing & switching, Forward and Reverse Proxy.
- Identity Access management, RADIUS, TACACS and Multifactor Authentication.
- Network access control (NAC), TCP/UDP, VLAN & IP Subnetting, DNS, SNMP, DHCP, VTP, Link Aggregation.
- Symmetric, Asymmetric encryption, Hashing algorithms, DH key exchange, SSL Handshake.
- SD-WAN, ECMP. Policy Routes.
- High Availability, Clustering and redundancy Protocols VRRP, FGCP, HSRP.
- Dynamic Routing Protocol RIP, EIGRP, OSPF.
- IPSEC, SSL VPN.
- Certificate, Certificate Authority and deep packet inspection.
- Email Security SMTP, IMAP, POP3, Anti-spam, Mx record, SPF, DKIM, DMARC.
- Endpoint Security – Bitdefender, CrowdStrike, Forticlient EMS.

- Application Load Balancer, Web Application Firewall.

Hands On Experience

- Fortigate firewalls.
- PaloAlto Firewalls.
- Fortimail /Fortimail Cloud.
- Fortimanager.
- PaloAlto Panorama.
- FortiAnalyzer.
- Forticlient EMS.
- FortiAuthenticator.
- FortiSandbox.
- FortiWLC/WLM.
- FortiNAC.
- FortiADC.
- FortiProxy.
- FortiWeb.
- PulseSecure PSA-3000.

Major Projects

- Confidential Government Sector Customer: Fortinet Security Implementation (Fortimail Cloud- Fortigate 601E- FortiSandbox- Fortimanager – FortiClientEMS -FortiADC- FortiNAC (visibility and profiling)- FortiSIEM VM instalation).
- Confidential Bank Customer: Successfully migrated checkpoint firewall to PaloAlto 3250.
- Confidential Government Sector Customer: successfully migrated Fortinet firewall to PaloAlto 3220 in 2 different sites.
- Confidential Government Sector Customer: Fortinet Security Implementation (4 FortiGate – 2 FortiSandbox – 4 Fortimail – Fortimanager – FortiProxy basic setup).
- Confidential Bank Customer: 6 FortiGate, FortiAnalyzer and FortiManager.
- Confidential University Customer: Fortigate 3000D, FortiAnalyzer & FortiAuthenticator.
- Confidential Government Sector Customer: 2 FortiGate 500E.
- Confidential Bank Customer: 2 FortiAuthenticator.
- Confidential Private sector Customer: implement DCFW fortigate 1101E.
- Confidential Bank Customer: FortiGate 1500D & FortiAnalyzer 200F.
- Confidential Bank Customer: FortiGate 300E.
- Confidential Private sector Customer: FortiAuthenticatorVM.
- Confidential Government Sector Customer: migrate PaloAlto 2050 to 3220.
- Confidential Private sector Customer: migrate ASA to FGT101F.
- Confidential Private sector Customer: Migrate Sophos firewall to Fortigate 101F.
- Confidential Private sector Customer: Implement FortiGate 80E.
- Confidential Private sector Customer: Implement FortiGate 50E.
- Confidential Private sector Customer: FortiGate 60E.

- Confidential Customer: 2 FortiNAC VM.
- Confidential Customer: basic installation and configuration of FortiWLC/FortiWLM/FortiAP.

Formal Education

- BSc in Computer Engineering – Kuwait University, College of Engineering & Petroleum, Kuwait.

Personal Details

- Date of Birth: 07/07/1994
- Gender, Marital: Male, Married
- Country of residence: UAE
- Nationality: Jordanian
- Driver's License: Holding valid Kuwait and Jordan Driving License
- Languages known: English, Arabic.
- Passport No: Q201154

DINESH LN RAO MERUVA

Abu Dhabi-UAE

Professional Certifications



- **CCIE SECURITY** | Cisco Certified Internetwork Expert Security
- **CCNP Routing and Switching** | Cisco Certified Network Professional Routing and Switching
- **CCNA Wireless** | Cisco Certified Network Associate Wireless
- **PCNSE7** | Palo Alto Networks Certified Network Security Engineer 7
- **PCNSE8** | Palo Alto Networks Certified Network Security Engineer 8
- **ACE 6.1** | Accredited Configuration Engineer | Palo Alto
- **ACE 7.0** | Accredited Configuration Engineer | Palo Alto
- **ACE 7.1** | Accredited Configuration Engineer | Palo Alto
- **ACE 8.0** | Accredited Configuration Engineer | Palo Alto

Technical Skills

- IP/Security Network Design, Configuration, and Operation
- Routing Protocols (RIP, IS-IS, IGRP, EIGRP, OSPF, BGP), advanced routing policies, filter-based forwarding, virtual routers and routing instances
- VLAN, Inter VLAN Routing, STP/RSTP/MSTP, VTP, Link Aggregation, Stacking, Virtual Chassis
- Redundancy Protocols VRRP, GLBP, HSRP, NSRP, JSRP)
- IPSEC, SSL, GRE, GRE over IPSEC, Juniper Auto Connect and Dynamic VPN, Cisco DMVPN
- Stateful/State less Firewall Policies/ACL, Screening, UTM (Antivirus, Anti-spam, Deep Inspection, Web filtering), Intrusion detection and prevention systems, Network access control (NAC)
- QoS, Multicast and MPLS layer3 VPN
- Layer 2 protocols and technologies (ATM, Frame Relay, POS, Ethernet)

Hands on Experience

- Cisco ISE.
- Cisco ASA
- Cisco FTD with FMC
- Fortinet Firewalls - (Fortigate – 50E, 100D, 500D, 800D, 1500D, 3000D)
- Cisco ESA – Email Security Appliance with SMA
- Cisco ISE and Cisco ACS.
- Palo Alto central Management system | Panorama | M 100
- Palo Alto PA 5050 | PA 5020 | PA 3020 | PA 500 | PA 200 | PA VM
- Cisco Routers (38xx, 28xx, 18xx)
- Cisco Switches (65xx, 35xx, 29xx)
- Cisco NGFW and ASA.
- HP Aruba Switches (29xx)
- Avaya Switches
- Fortigate Firewalls | Forti Analyzer

Major Projects

- Unnamed | Abu Dhabi | Deployed Fortinet Firewalls connecting all the base camps of Unnamed.
- Higher College of Technology | Abu Dhabi, Dubai, RAK | | Deployed PaloAlto Firewalls for a large Campus Network.
- Advanced Military Maintenance, Repair and Overhaul Center (AMMROC) | Abu Dhabi | Deployed DR Palo Alto setup. Integrated DC and DR firewalls to Panorama.
- Executive Affairs Authority | Abu Dhabi | Deployed Palo Alto in DC and DR.
- National Drilling Company (ADNOC Drilling) | Abu Dhabi | Deployed Palo Alto in the Head Quarters connecting all the branches of ADNOC Drilling.
- ESNAAD (ADNOC Logistics & Services) | Abu Dhabi | Deployed Forti Analyzer
- Health Authority | Abu Dhabi | Deployed Core Firewall in Head Quarters | Fortinet
- Emirates Driving Company | Abu Dhabi | Deployed Fortinet firewall in DC and DR.
- Rotana Hotels and Resorts | Abu Dhabi | Deployed Palo Alto Firewalls throughout the properties of Rotana and acts as Primary technical contact for Firewalls.

SALEEM POTTACHOLA

Abu Dhabi-UAE

Professional Certifications



- Cisco Certified Internetwork Expert – CCIE Collaboration# 51113
- Cisco Certified Network Professional - Voice (CCNP - Voice), Cisco ID: CSC011287996
- Cisco Certified Network Associate - Voice (CCNA-Voice), Cisco ID: CSC011287996
- Cisco Certified Network Associate – Routing and Switching (CCNA), Cisco ID: CSC011287996
- Cisco Collaboration Field Engineer Representative 1
- Express Foundation Systems Engineer Representative
- Express Collaboration Systems Engineer Representative v2
- Cisco WebEx Solutions Design and Implementation 648-232
- Designing and Implementing Cisco Unified Communications on Unified Computing Systems 648-244

Technical Skills

- Cisco Telephony - Call Manager and IM and Presence
- Cisco Voice Mail – Unity connection
- Cisco Voice Gateways – 4000 (4th Gen), 3900, 2900 and 5400 series routers
- Cisco Contact Center Enterprise - ICM, CVP, CUSP, PGW, CUIC and Finesse
- Cisco Contact Center Express, IVR, reporting and recording.
- Cisco Telepresence
- Cisco Wireless – 8510, 5520, 2504 controller, Mobility Express
- Cisco Prime Infrastructure for wired and wireless network
- Cisco WIPS
- Cisco ACS (Access Control Server) as a TACACS+ Server.
- Cisco ACS (Access Control Server) as a RADIUS Server.
- Cisco Network - Switches and routers

Hands on Experience

- Cisco Unified Communication Manager – 6.x, 7.x, 9.x, 11.x
- Cisco Unified Contact Center Enterprise – 7.x, 9.x
- Cisco Unified Contact Center Express – 11.x
- Cisco Unity Connection 11.x
- Cisco IM and Presence 11.x
- Cisco Unified Intelligence Center Reporting – 9.x 11.x
- Recording – Imagicle, Cisco Media sense
- Cisco CUBE
- Cisco UCS Manager
- Cisco Wireless Controller – 8.x on 8510, 5520, 2504

- Cisco Mobility Express – 8.x
- Cisco Prime Infrastructure 3.1
- Cisco MSE for WIPS 8.x
- Cisco ACS for TACACS+ and Radius – 5.x
- Cisco Routers (39xx, 29xx, 54xx)
- Cisco Switches (35xx, 29xx)

Major Projects

- Confidential: Cisco Wireless Infrastructure deployment. 2 sites with 1000 AP's, Cisco ACS for TACACS+ and Radius, Cisco Prime for wireless and wired monitoring and device management. Site survey
- Kings College Hospital: IT Infrastructure Project - Datacenter, IP Telephony, Wired and wireless network, Contact Center Express design and deployment. BE6000, CUCM 11.5, WLC 5520, UCCX 11.x, Switches 3850, 2960, Routers 29xx
- MOE – IT Infrastructure Project – Wired and Wireless network for Hatta school
- Yas Mall – IP Telephony recording – Imagic recording for CUCM 11.x
- Borouge: Cisco WebEx Meeting server (CWMS) on Premise
- Confidential – SX80 codec, speaker track for 20 sites across UAE
- ADHA – Cisco WIPS with Cisco MSC and Cisco Prime
- Confidential – Cisco Telepresence content Server (TCS) for Telepresence video conference recording
- Aldar Academy – Aruba wireless site survey and heat map
- Al Dhafra Sport Club – IT Infrastructure - Cisco wireless and wired network deployment

THANK YOU